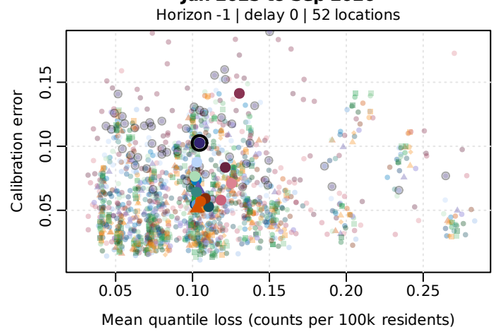
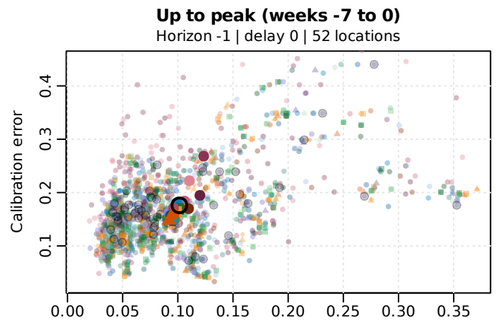
covid | all 52 locations | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

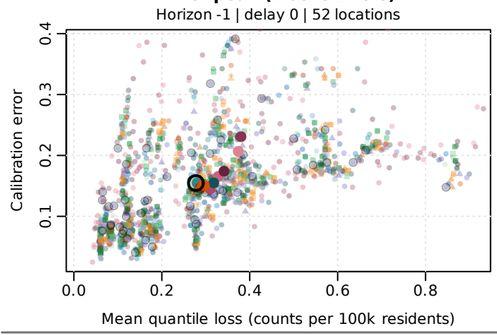
Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.
Small: single locations; large: means over locations; axes: 1st to 99th percentile of both.

Horizon -1

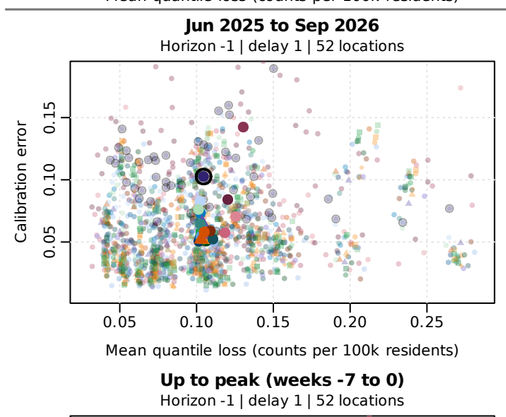
Feedback delay 0

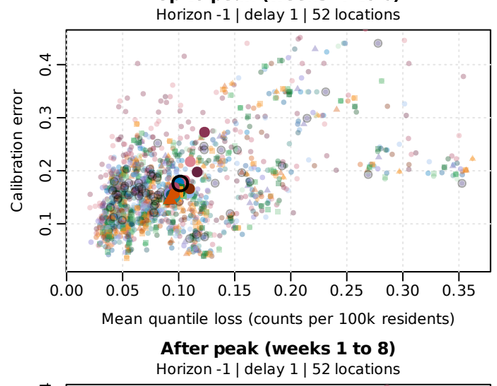


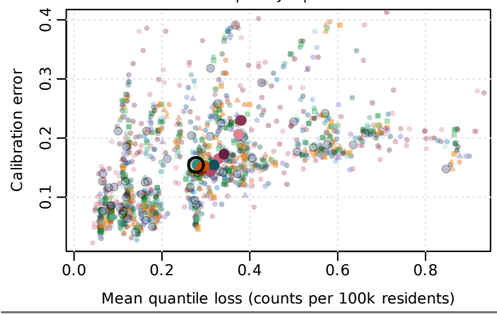




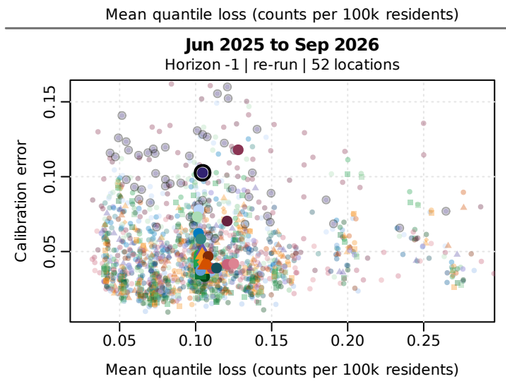
Feedback delay 1

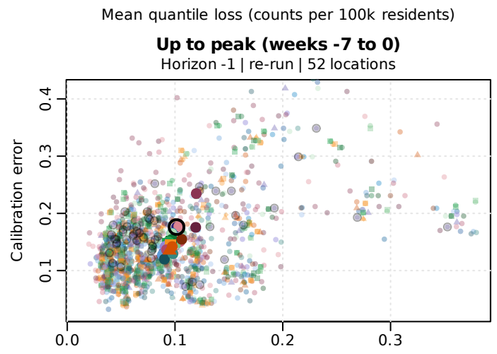


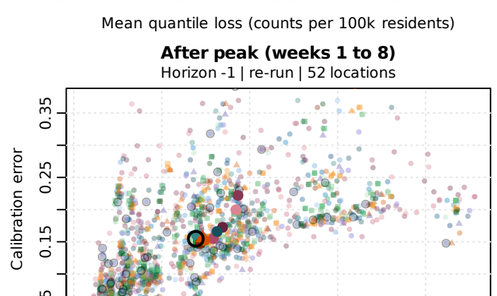




delay 0 + revisions (re-run)

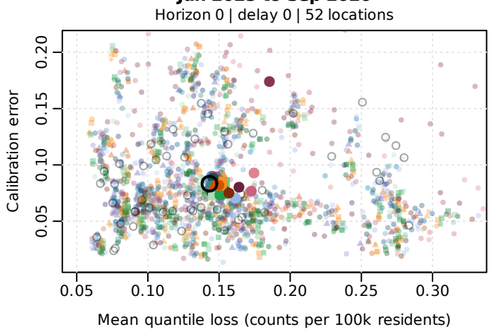


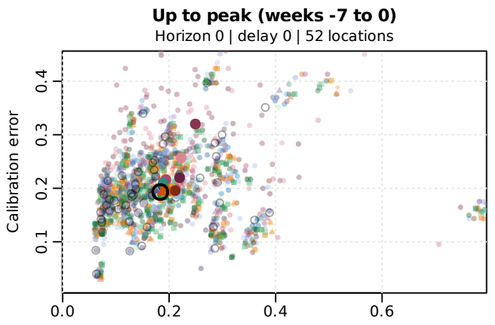


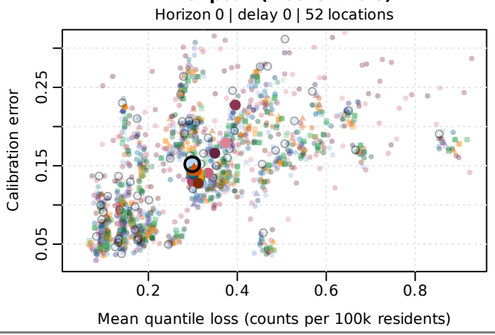


Horizon 0

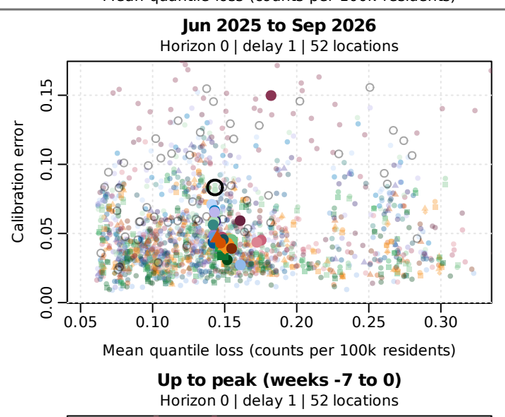
Feedback delay 0

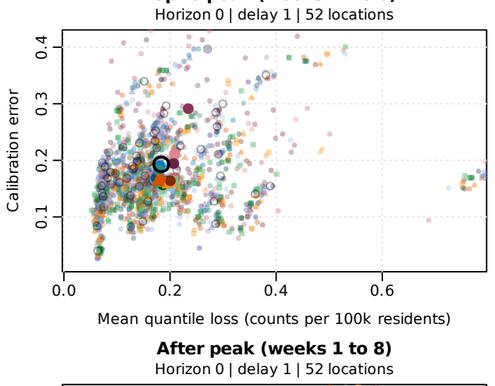


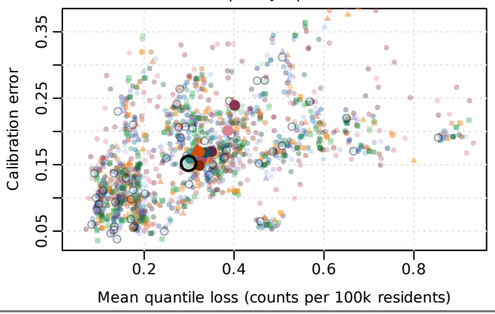




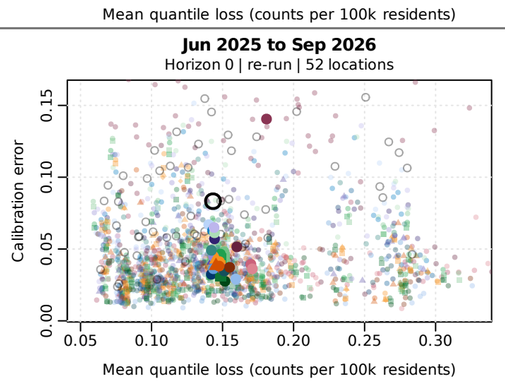
Feedback delay 1

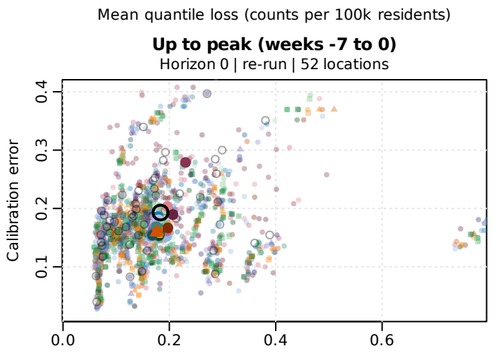


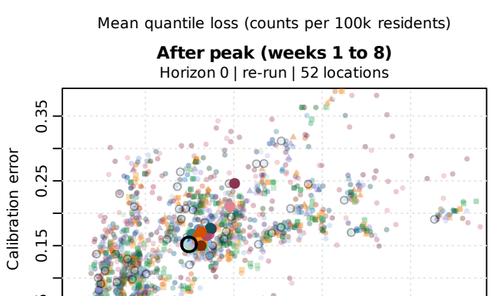




delay 0 + revisions (re-run)

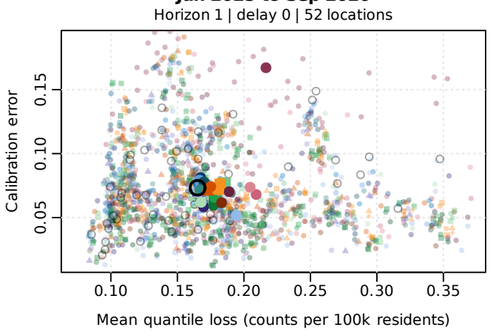


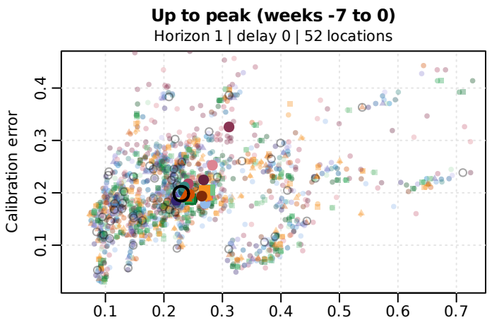


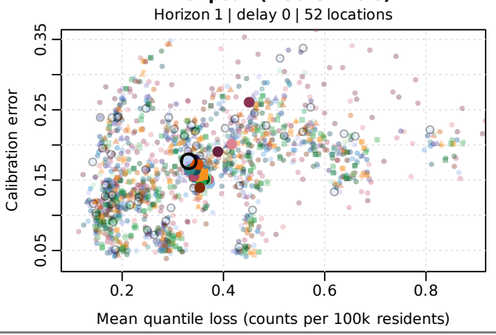


Horizon 1

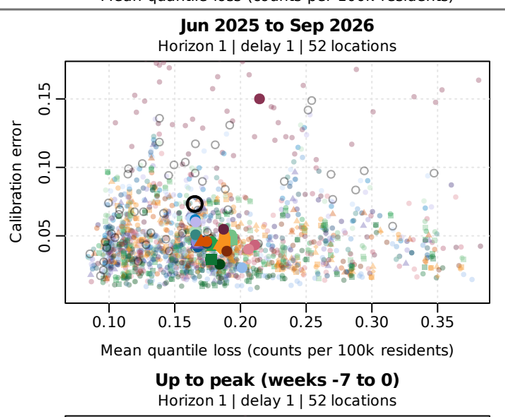
Feedback delay 0

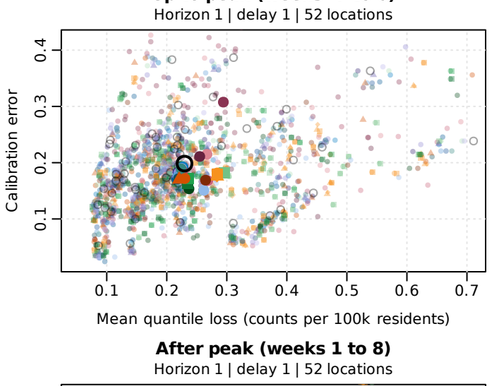


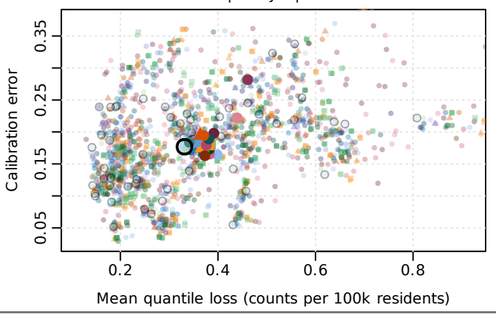




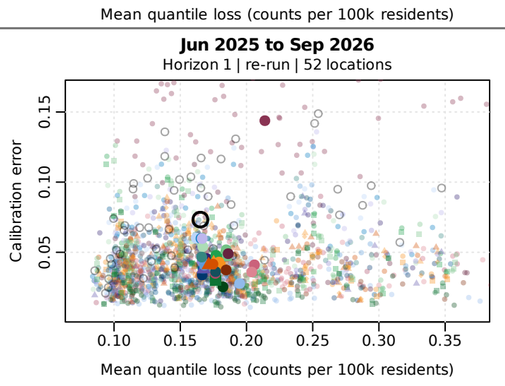
Feedback delay 1

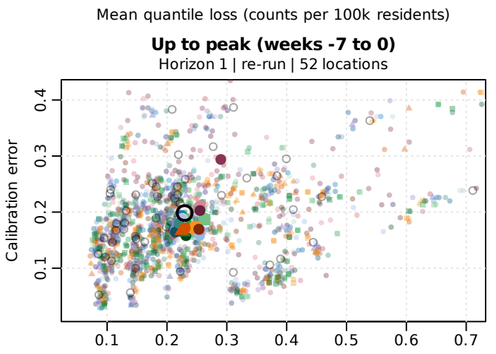


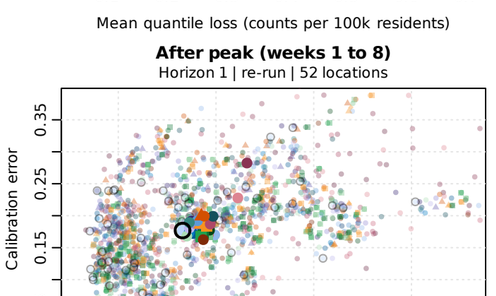




delay 0 + revisions (re-run)

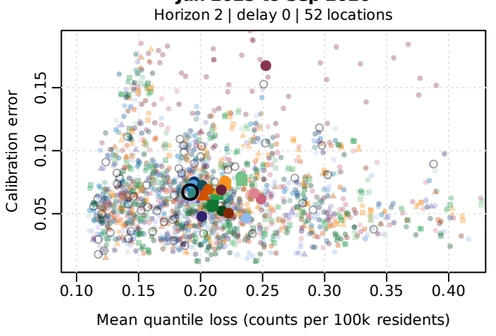


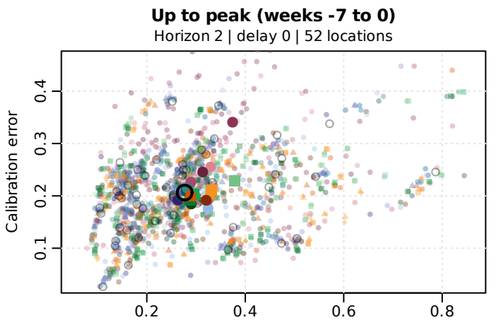


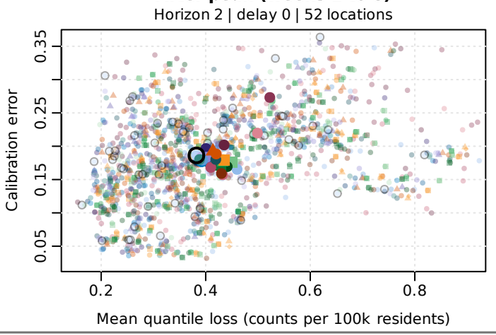


Horizon 2

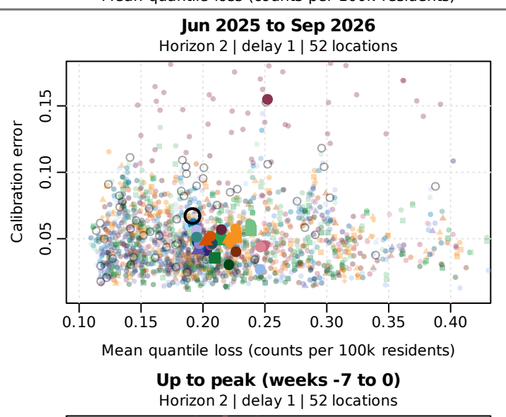
Feedback delay 0

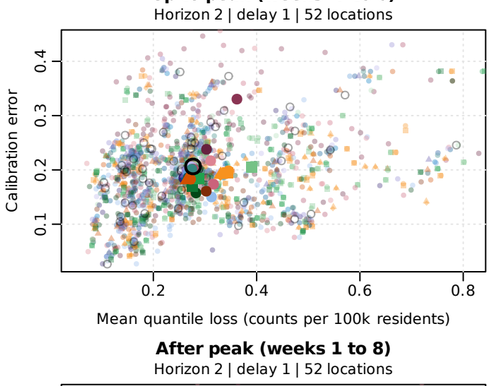


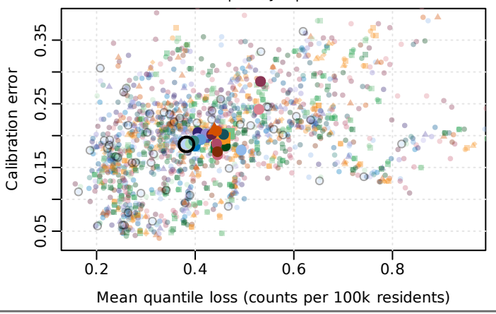




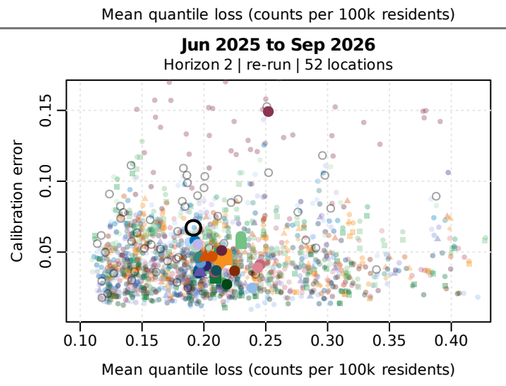
Feedback delay 1

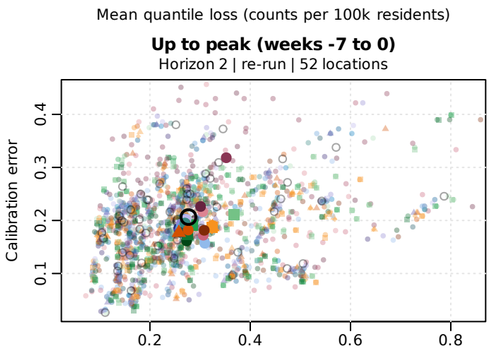


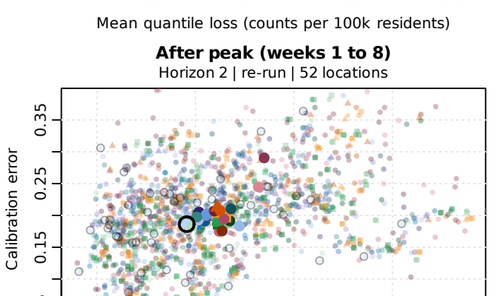




delay 0 + revisions (re-run)

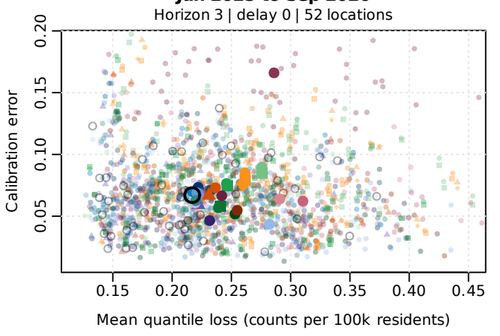


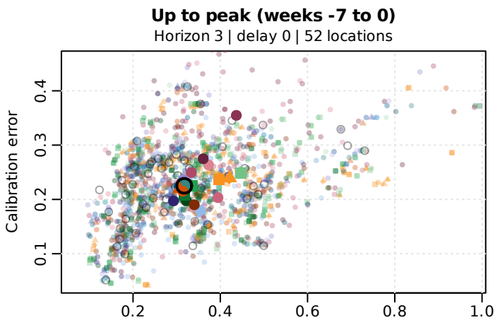


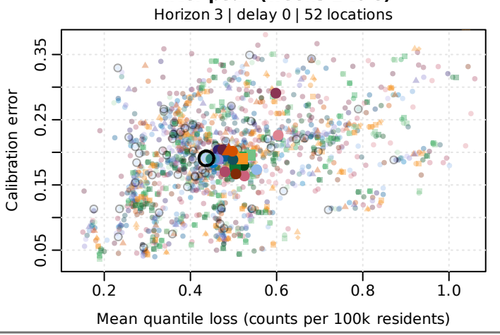


Horizon 3

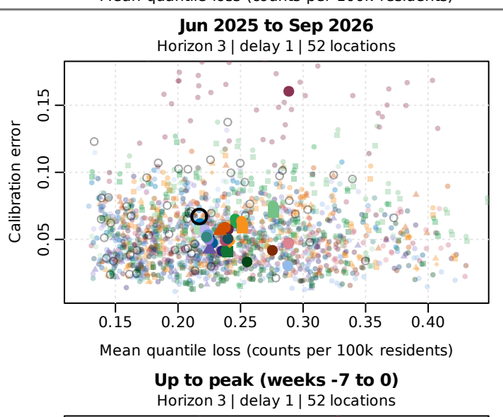
Feedback delay 0

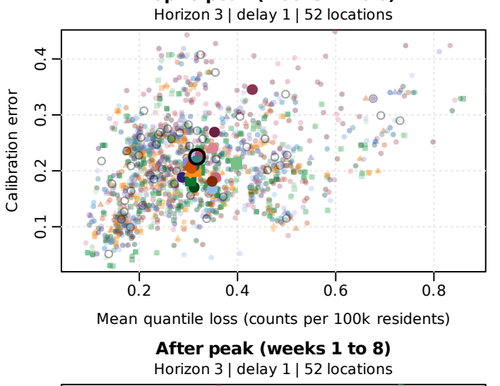


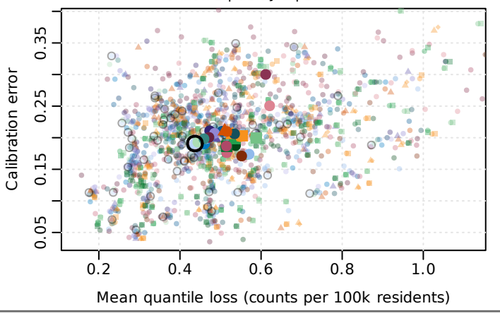




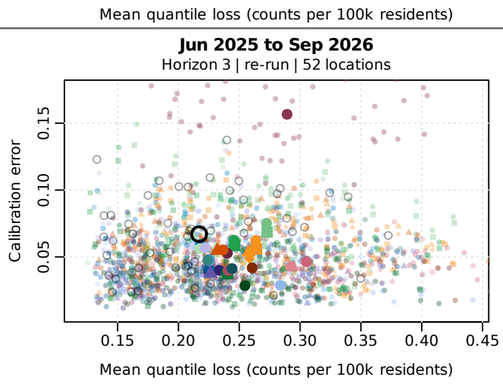
Feedback delay 1

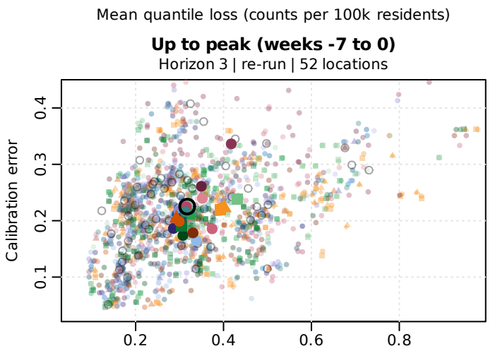


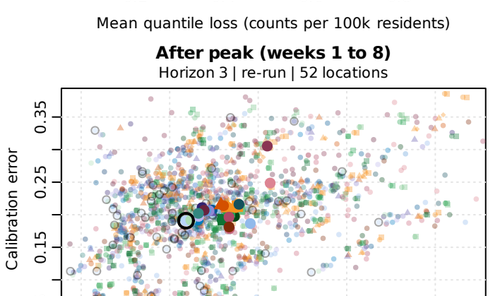




delay 0 + revisions (re-run)







Single trackers
- Original
- Fixed MultiQT (eta=0.005*S0)
- Leaky MultiQT (eta=0.008*S0, leak 0.2)
- Delphi REF-op (sqrt, eta=0.03*Q90 of 20, slow 0.003, leak 0.1)
- Delphi flu prod (sqrt per 100k, eta=0.018)
- Count-slope rate (min m=0.1)
- Target oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=32)
- Rate grid: learned, no sharing, common weights (H=32)
- Rate grid: learned + sharing (H=32)
- Rate grid: learned + sharing, common weights (H=16)
- Rate grid: sharing + outer (H=32, outer m=0.001)
- Rate grid: sharing, common weights + outer (H=16, outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, H=32)
- Starts x rate grid: learned (H=32)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (H=32, outer m=0.001)
- Starts x rate grid + outer (H=32, outer m=0.001)

Score-based experts
- Decline MultiQT (eta=0.05*S0, declining context, delta=0)
- Score experts: equal weights
- Score experts: learned weights per level (H=32)
- Score experts: learned, (1 - G) x mixture + G x original (H=32, delta=0)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.03)
- Studentized median (RMS of 5, median start, eta=0.03)
- Studentized median (base 80% width, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, median start, eta=0.1)

Studentized full base
- Studentized full base (base 80% width, eta=0.1)
- Studentized full base (level RMS of 5, eta=0.03)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Not drawn (identical forecasts): Decline MultiQT (eta=0.05*S0, declining context, delta=0) = Original; Score experts: learned, (1 - G) x mixture + G x original (H=32, delta=0) = Score experts: learned weights per level (H=32)
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base. Median start: all quantiles start at the median (offsets 0).
Studentized full base: offsets start at the base; level RMS = the RMS of each level's last 5 released residuals (weighted isotonic projection).

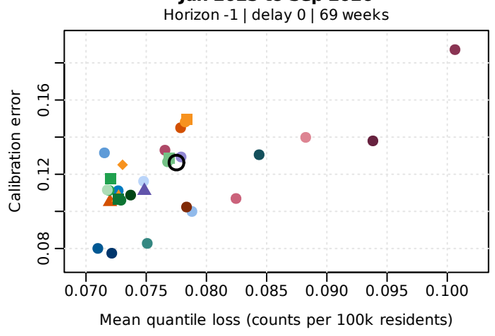
covid | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

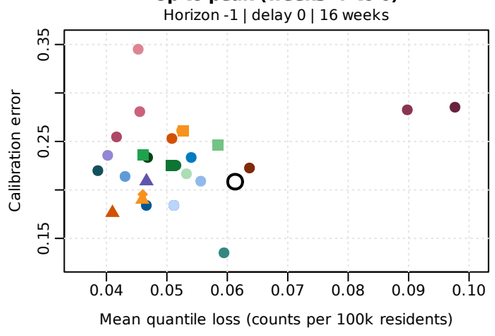
Horizon -1

Feedback delay 0

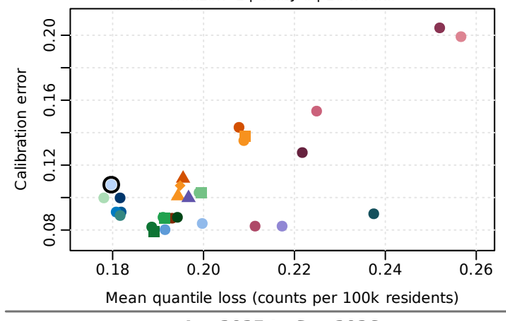
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 16 weeks

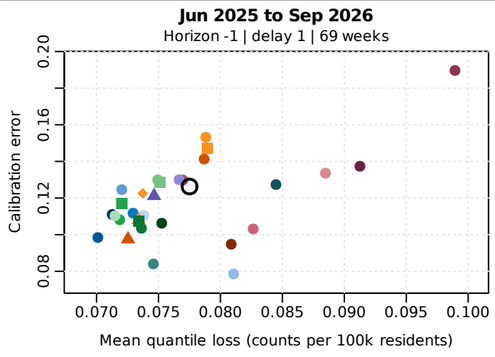


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 16 weeks

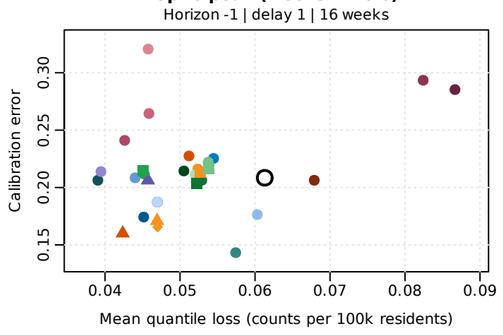


Feedback delay 1

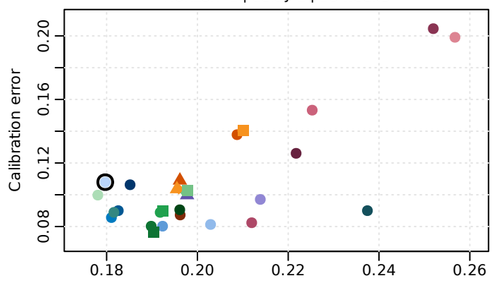
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 16 weeks

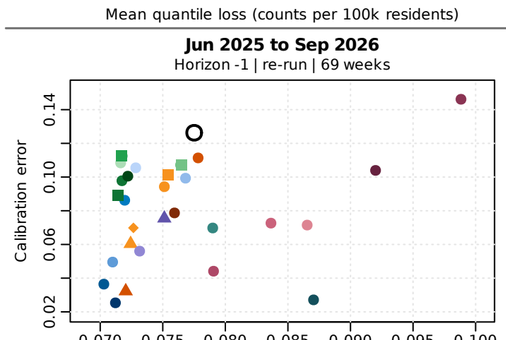


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 16 weeks

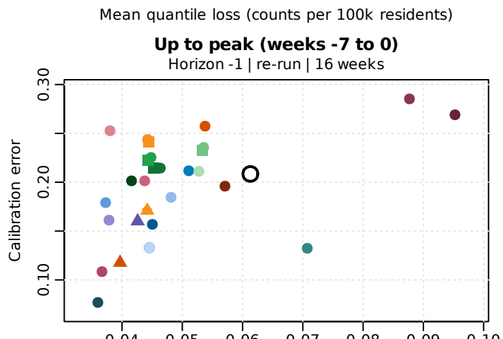


delay 0 + revisions (re-run)

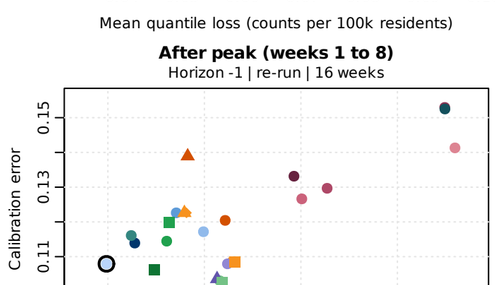
Jun 2025 to Sep 2026
Horizon -1 | re-run | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 16 weeks



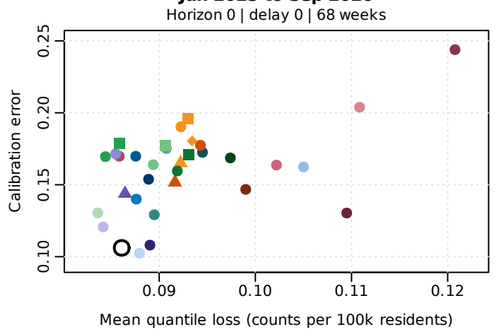
After peak (weeks 1 to 8)
Horizon -1 | re-run | 16 weeks



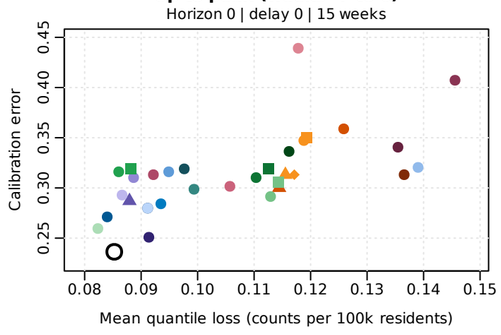
Horizon 0

Feedback delay 0

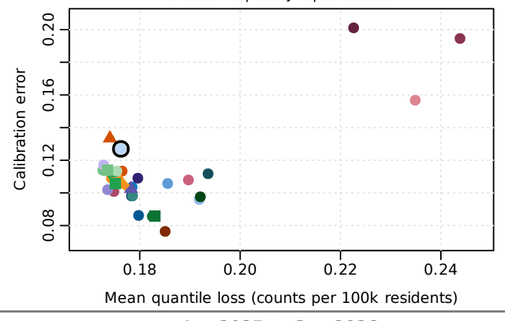
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 15 weeks

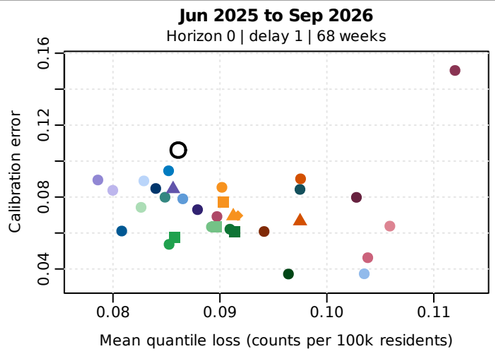


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 16 weeks

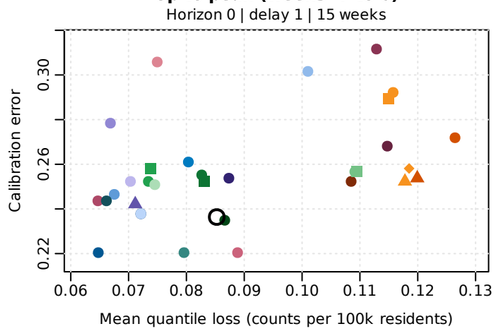


Feedback delay 1

Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 68 weeks



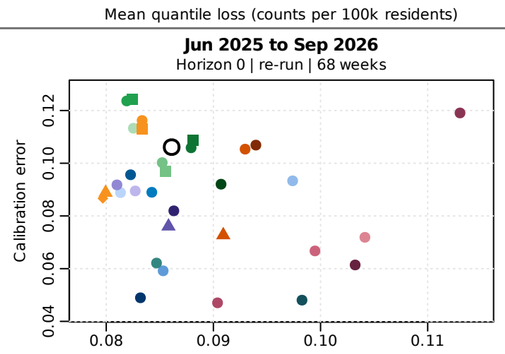
Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 15 weeks



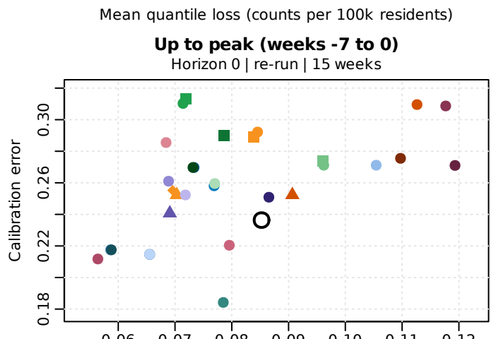
After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 16 weeks

delay 0 + revisions (re-run)

Jun 2025 to Sep 2026
Horizon 0 | re-run | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 15 weeks

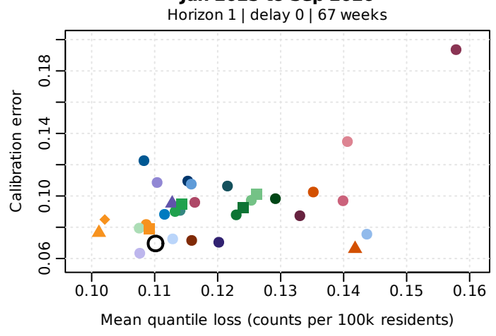


After peak (weeks 1 to 8)
Horizon 0 | re-run | 16 weeks

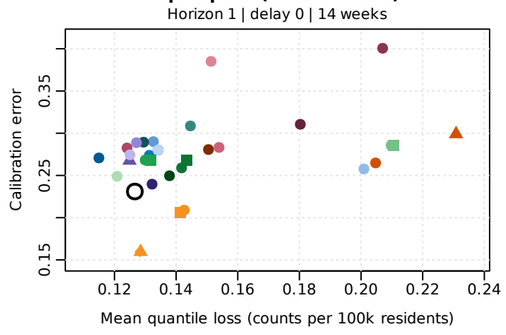
Horizon 1

Feedback delay 0

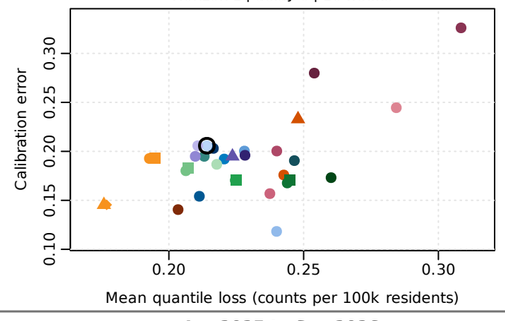
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 14 weeks

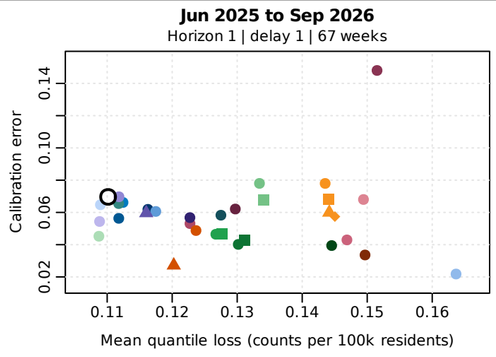


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 16 weeks

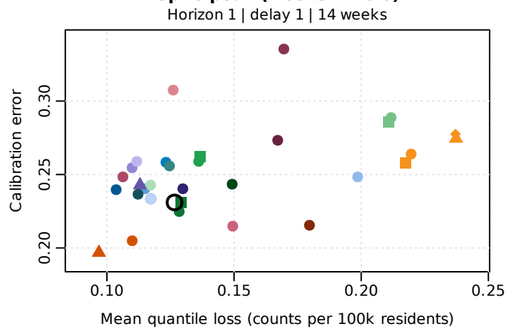


Feedback delay 1

Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 67 weeks



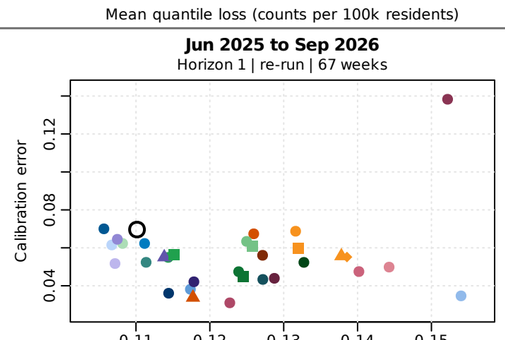
Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 14 weeks



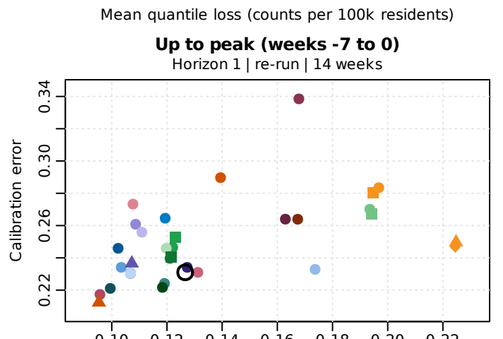
After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 16 weeks

delay 0 + revisions (re-run)

Jun 2025 to Sep 2026
Horizon 1 | re-run | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 14 weeks

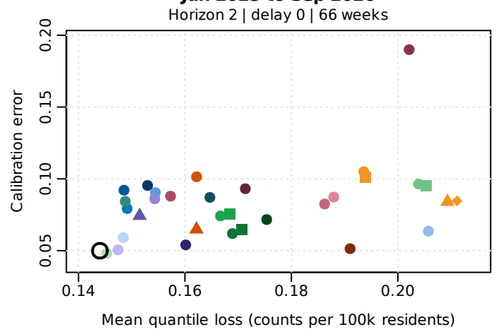


After peak (weeks 1 to 8)
Horizon 1 | re-run | 16 weeks

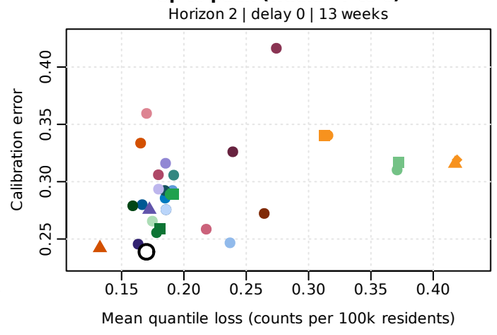
Horizon 2

Feedback delay 0

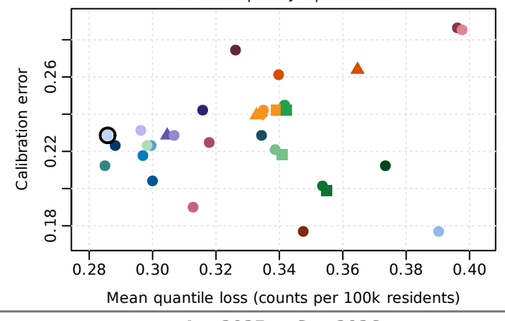
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 13 weeks

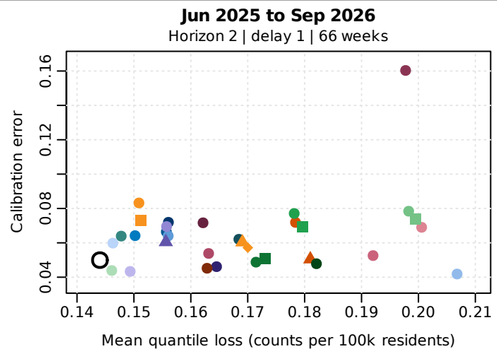


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 16 weeks

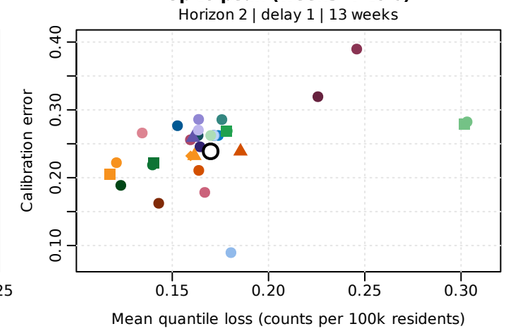


Feedback delay 1

Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 66 weeks



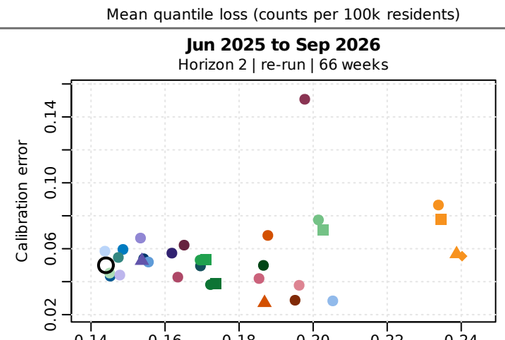
Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 13 weeks



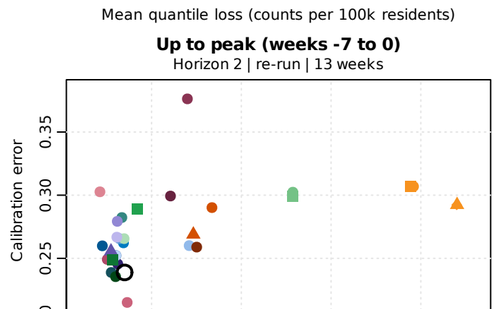
After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 16 weeks

delay 0 + revisions (re-run)

Jun 2025 to Sep 2026
Horizon 2 | re-run | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 13 weeks

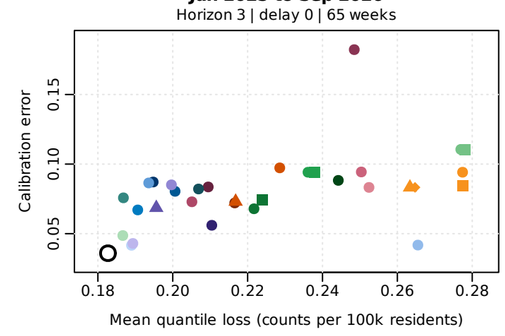


After peak (weeks 1 to 8)
Horizon 2 | re-run | 16 weeks

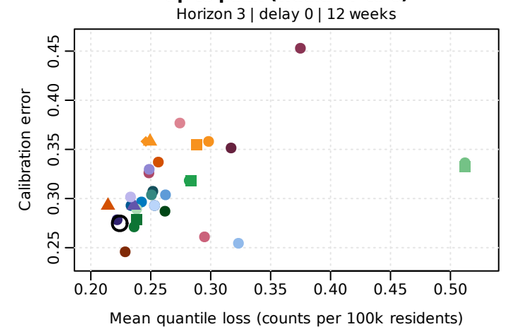
Horizon 3

Feedback delay 0

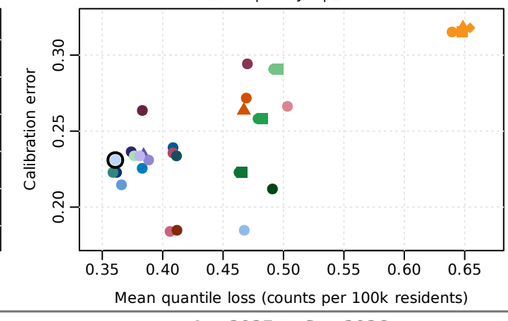
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 12 weeks

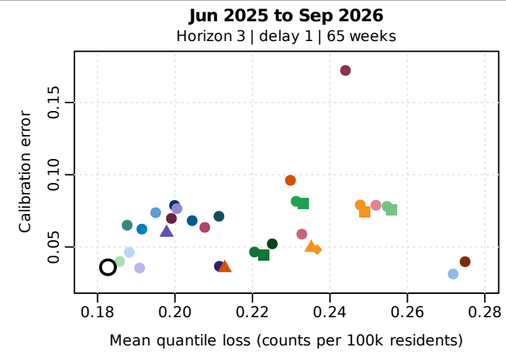


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 16 weeks

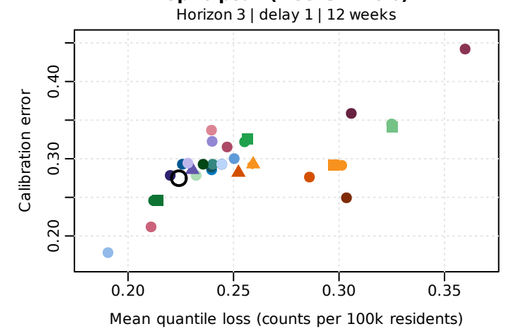


Feedback delay 1

Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 65 weeks



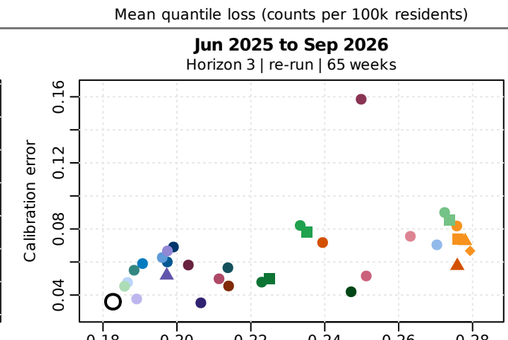
Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 12 weeks



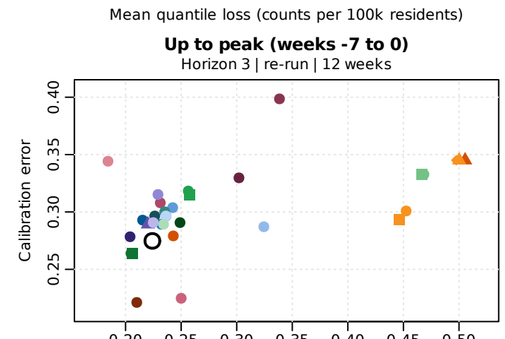
After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 16 weeks

delay 0 + revisions (re-run)

Jun 2025 to Sep 2026
Horizon 3 | re-run | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 12 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 16 weeks

Single trackers
- Original
- Fixed MultiQT (eta=0.005*S0)
- Leaky MultiQT (eta=0.008*S0, leak 0.2)
- Delphi REF-op (sqrt, eta=0.03*Q90 of 20, slow 0.003, leak 0.1)
- Delphi flu prod (sqrt per 100k, eta=0.018)
- Count-slope rate (min m=0.1)
- Target oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=32)
- Rate grid: learned, no sharing, common weights (H=32)
- Rate grid: learned + sharing (H=32)
- Rate grid: learned + sharing, common weights (H=16)
- Rate grid: sharing + outer (H=32, outer m=0.001)
- Rate grid: sharing, common weights + outer (H=16, outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, H=32)
- Starts x rate grid: learned (H=32)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (H=32, outer m=0.001)
- Starts x rate grid + outer (H=32, outer m=0.001)

Score-based experts
- Decline MultiQT (eta=0.05*S0, declining context, delta=0)
- Score experts: equal weights
- Score experts: learned weights per level (H=32)
- Score experts: learned, (1 - G) x mixture + G x original (H=32, delta=0)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.03)
- Studentized median (RMS of 5, median start, eta=0.03)
- Studentized median (base 80% width, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, median start, eta=0.1)

Studentized full base
- Studentized full base (base 80% width, eta=0.1)
- Studentized full base (level RMS of 5, eta=0.03)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Not drawn (identical forecasts): Decline MultiQT (eta=0.05*S0, declining context, delta=0) = Original; Score experts: learned, (1 - G) x mixture + G x original (H=32, delta=0) = Score experts: learned weights per level (H=32)
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base. Median start: all quantiles start at the median (offsets 0).
Studentized full base: offsets start at the base; level RMS = the RMS of each level's last 5 released residuals (weighted isotonic projection).

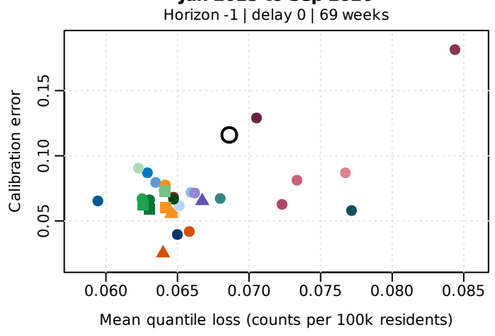
covid | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

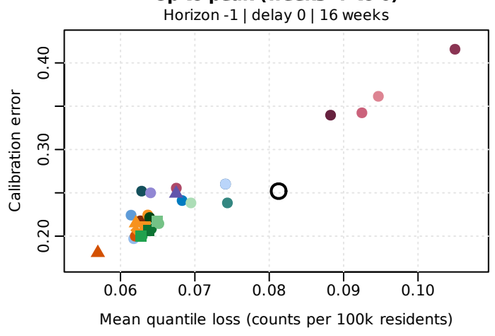
Horizon -1

Feedback delay 0

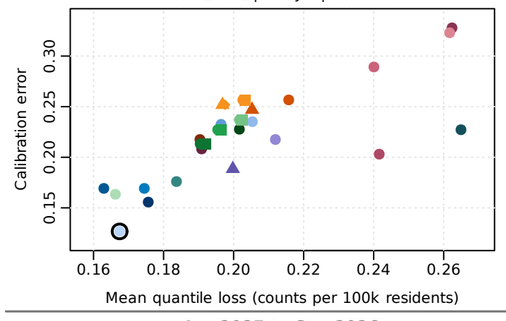
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 16 weeks

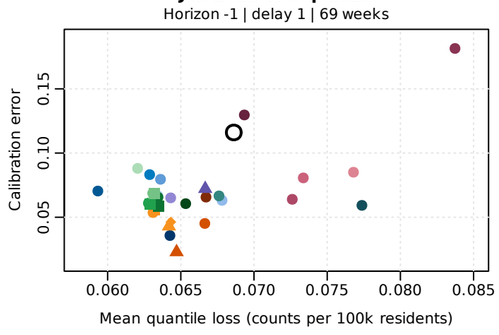


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 9 weeks

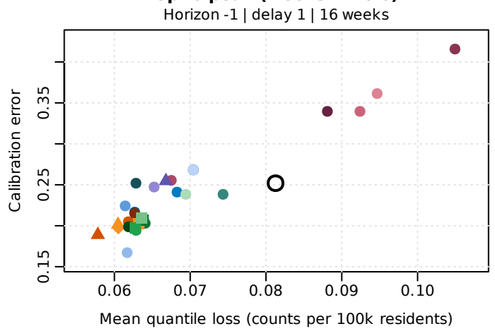


Feedback delay 1

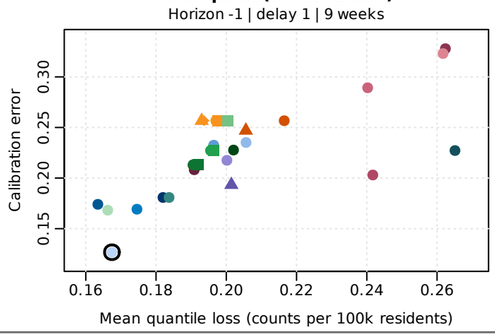
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 16 weeks

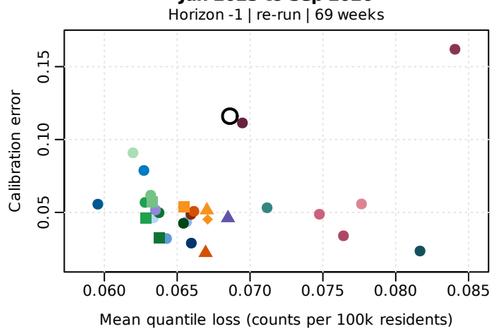


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 9 weeks

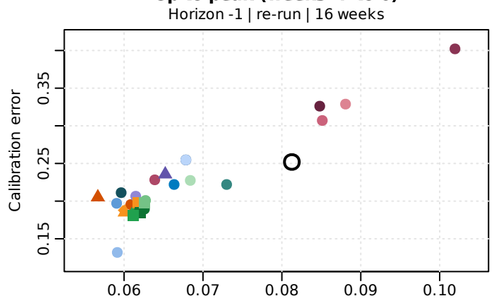


delay 0 + revisions (re-run)

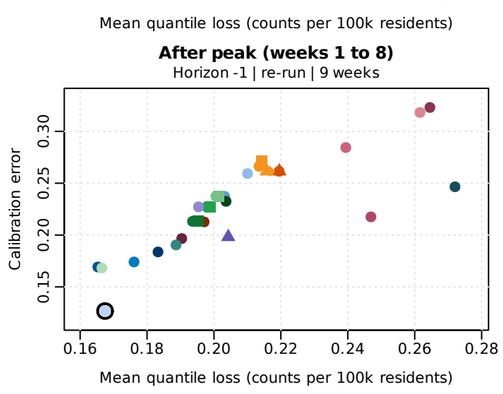
Jun 2025 to Sep 2026
Horizon -1 | re-run | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 16 weeks



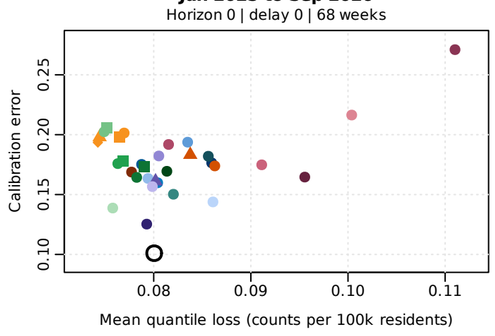
After peak (weeks 1 to 8)
Horizon -1 | re-run | 9 weeks



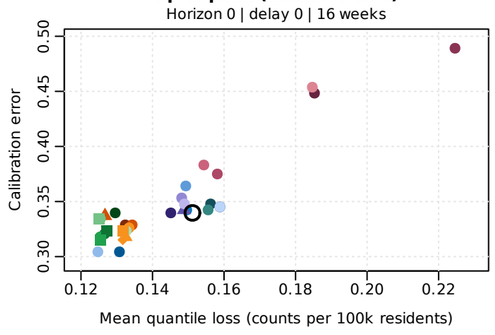
Horizon 0

Feedback delay 0

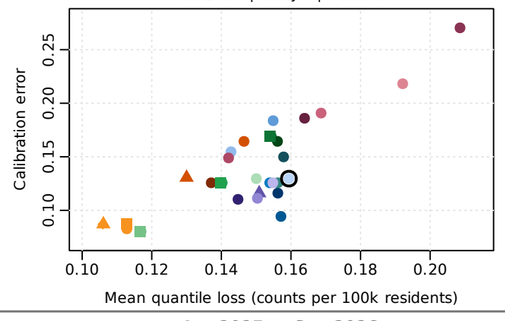
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 16 weeks

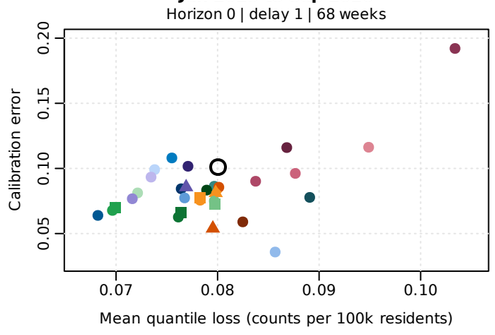


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 9 weeks

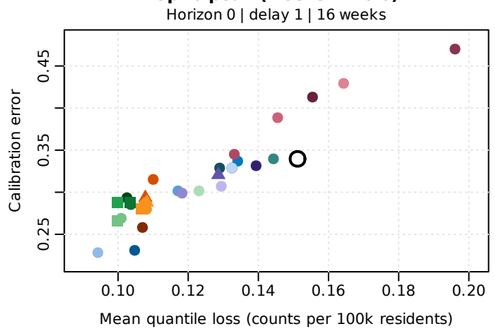


Feedback delay 1

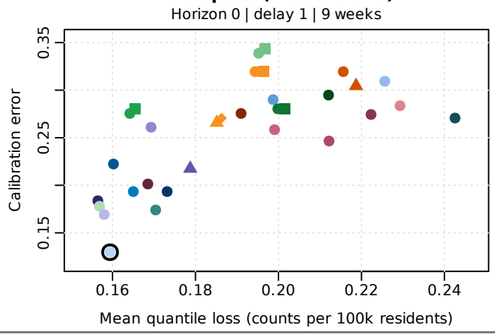
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 16 weeks

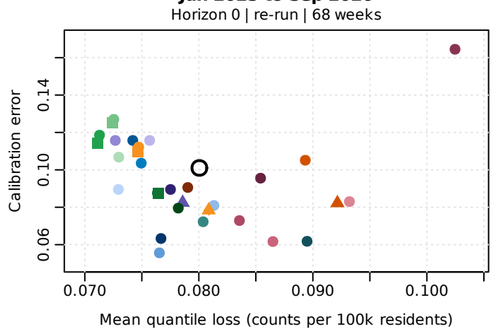


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 9 weeks

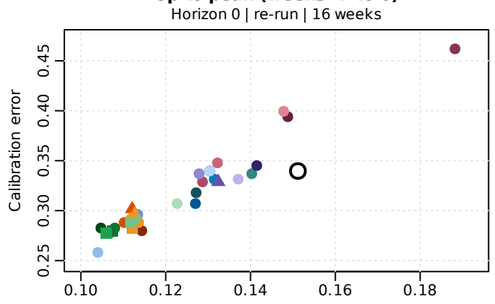


delay 0 + revisions (re-run)

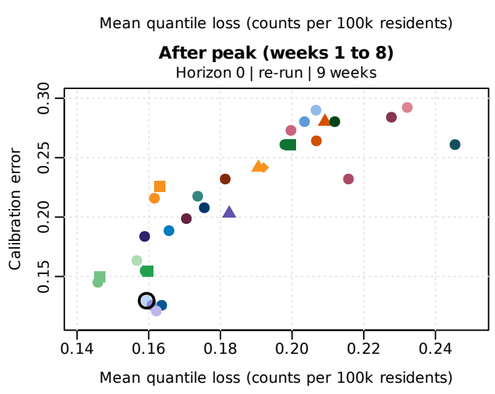
Jun 2025 to Sep 2026
Horizon 0 | re-run | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 16 weeks



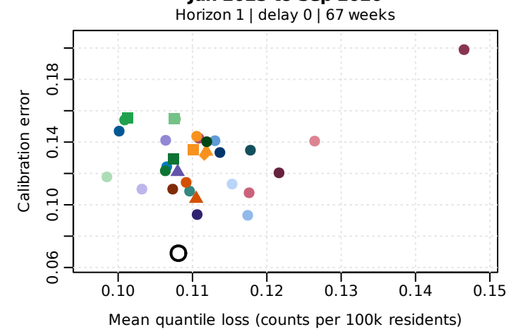
After peak (weeks 1 to 8)
Horizon 0 | re-run | 9 weeks



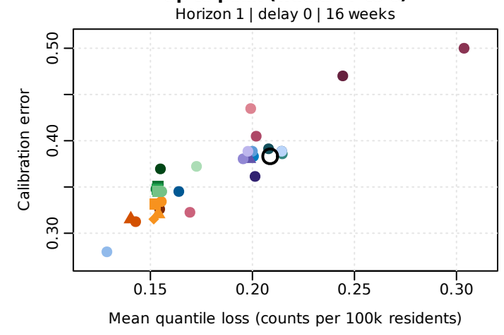
Horizon 1

Feedback delay 0

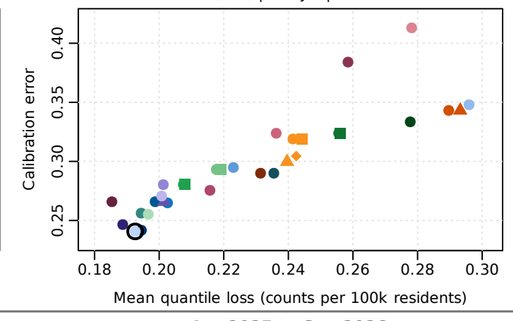
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 16 weeks

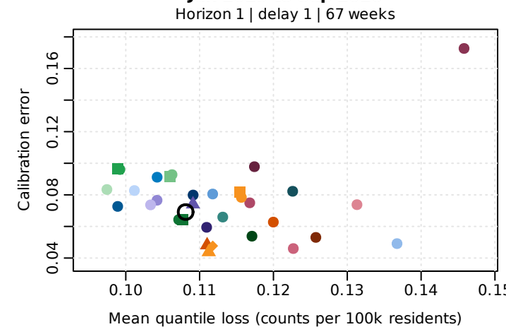


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 9 weeks

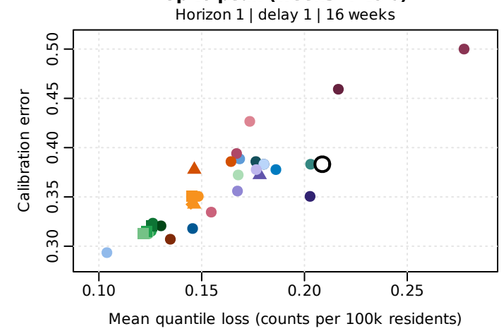


Feedback delay 1

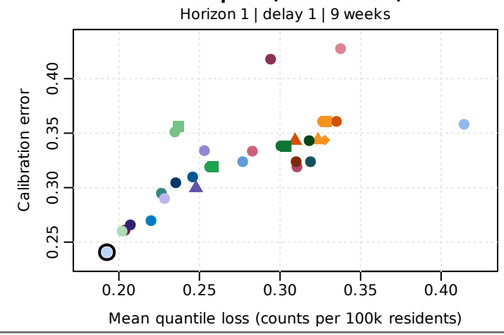
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 16 weeks

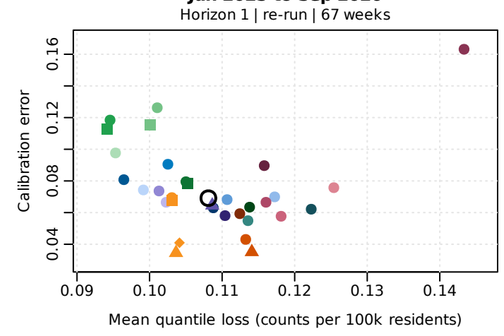


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 9 weeks

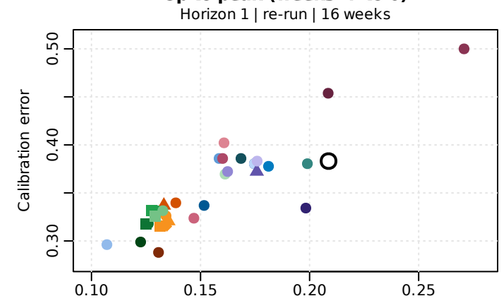


delay 0 + revisions (re-run)

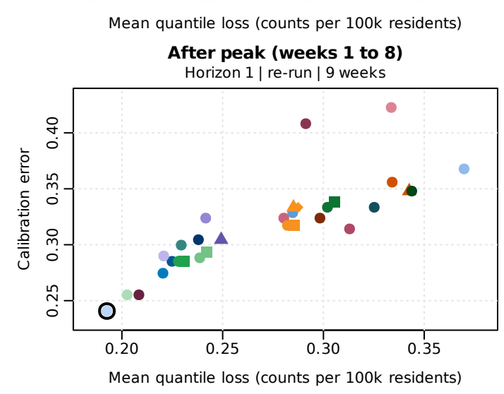
Jun 2025 to Sep 2026
Horizon 1 | re-run | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 16 weeks



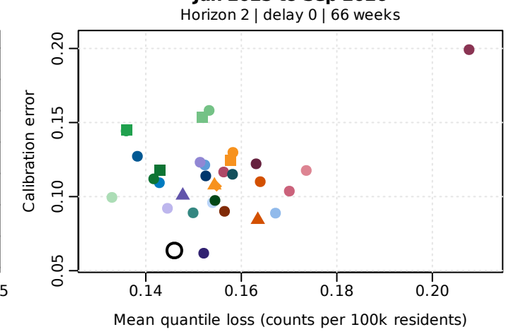
After peak (weeks 1 to 8)
Horizon 1 | re-run | 9 weeks



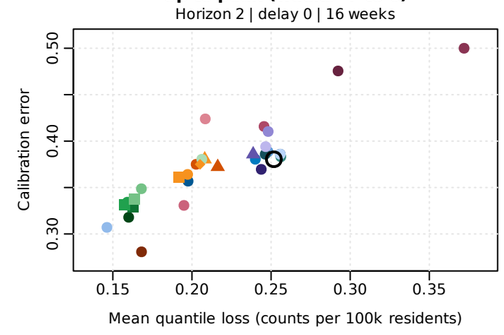
Horizon 2

Feedback delay 0

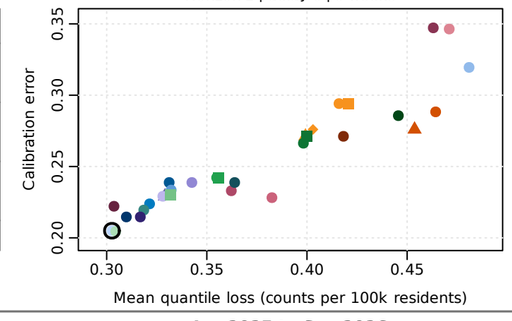
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 16 weeks

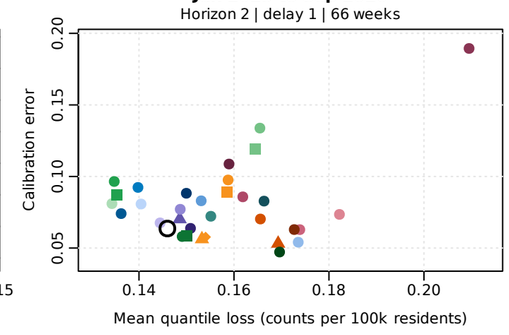


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 9 weeks

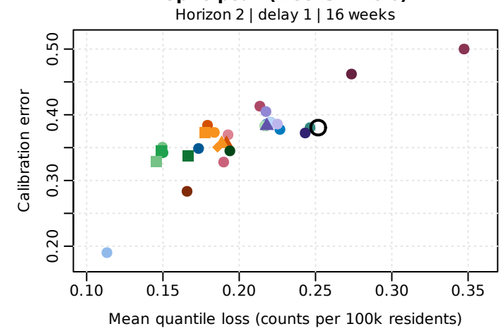


Feedback delay 1

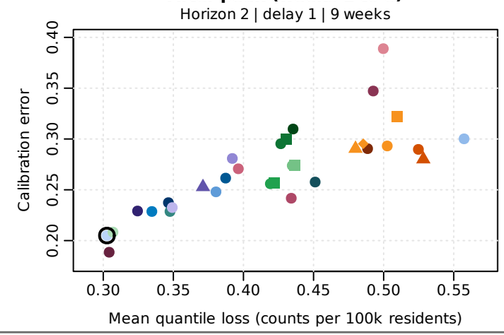
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 16 weeks

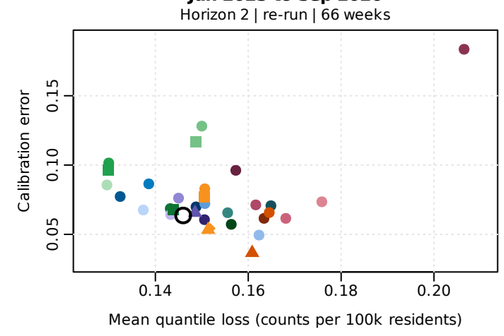


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 9 weeks

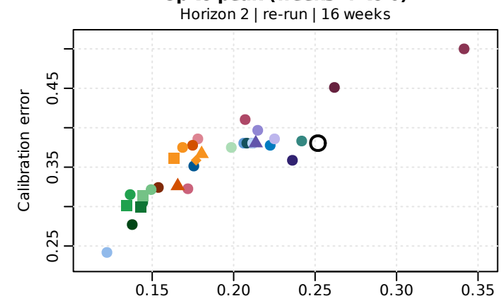


delay 0 + revisions (re-run)

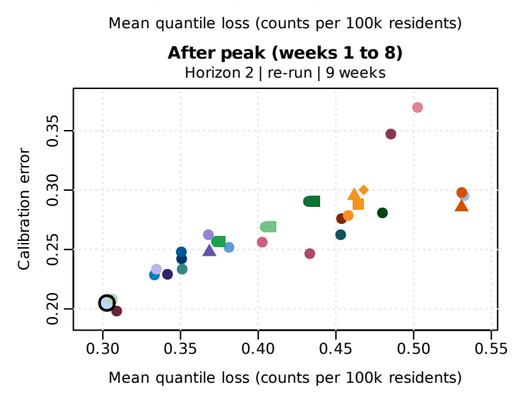
Jun 2025 to Sep 2026
Horizon 2 | re-run | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 16 weeks



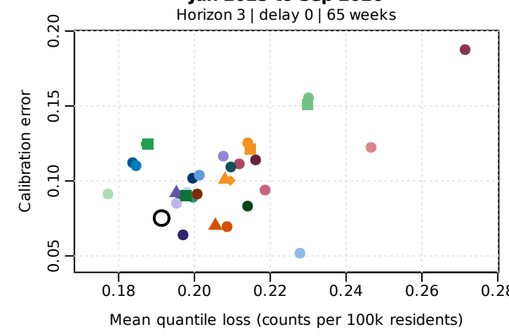
After peak (weeks 1 to 8)
Horizon 2 | re-run | 9 weeks



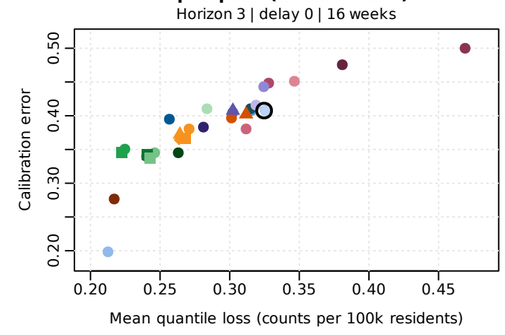
Horizon 3

Feedback delay 0

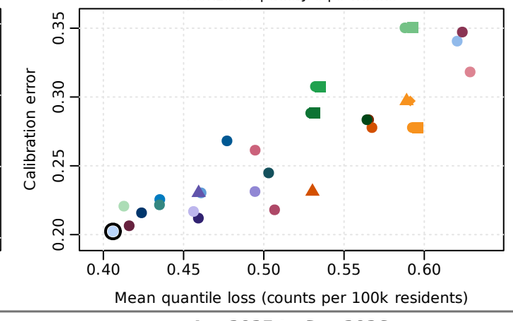
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 16 weeks

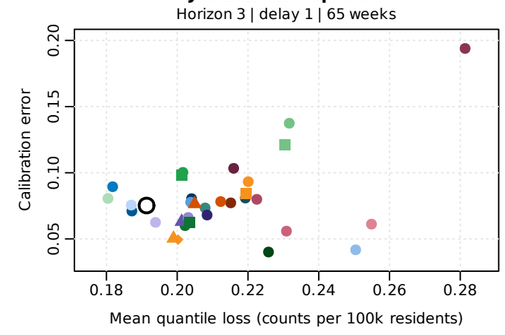


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 9 weeks

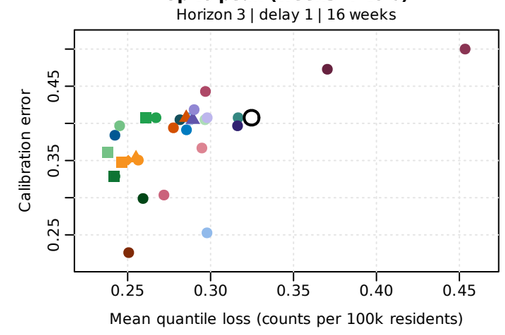


Feedback delay 1

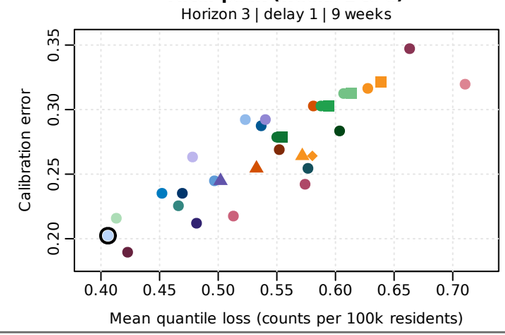
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 16 weeks

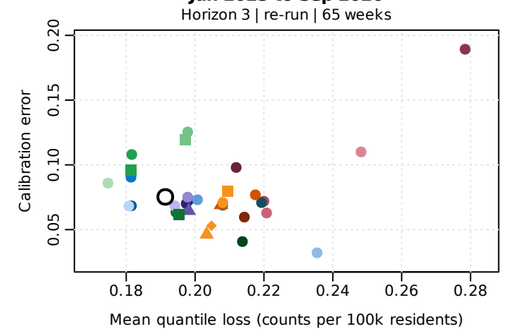


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 9 weeks

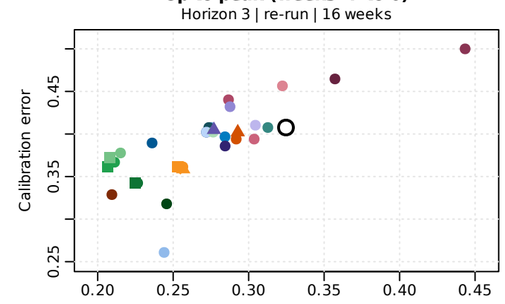


delay 0 + revisions (re-run)

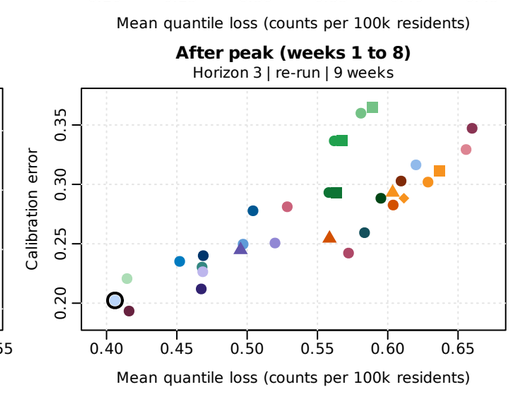
Jun 2025 to Sep 2026
Horizon 3 | re-run | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 16 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 9 weeks



Single trackers
- Original
- Fixed MultiQT (eta=0.005*S0)
- Leaky MultiQT (eta=0.008*S0, leak 0.2)
- Delphi REF-op (sqrt, eta=0.03*Q90 of 20, slow 0.003, leak 0.1)
- Delphi flu prod (sqrt per 100k, eta=0.018)
- Count-slope rate (min m=0.1)
- Target oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=32)
- Rate grid: learned, no sharing, common weights (H=32)
- Rate grid: learned + sharing (H=32)
- Rate grid: learned + sharing, common weights (H=16)
- Rate grid: sharing + outer (H=32, outer m=0.001)
- Rate grid: sharing, common weights + outer (H=16, outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, H=32)
- Starts x rate grid: learned (H=32)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (H=32, outer m=0.001)
- Starts x rate grid + outer (H=32, outer m=0.001)

Score-based experts
- Decline MultiQT (eta=0.05*S0, declining context, delta=0)
- Score experts: equal weights
- Score experts: learned weights per level (H=32)
- Score experts: learned, (1 - G) x mixture + G x original (H=32, delta=0)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.03)
- Studentized median (RMS of 5, median start, eta=0.03)
- Studentized median (base 80% width, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, median start, eta=0.1)

Studentized full base
- Studentized full base (base 80% width, eta=0.1)
- Studentized full base (level RMS of 5, eta=0.03)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Not drawn (identical forecasts): Decline MultiQT (eta=0.05*S0, declining context, delta=0) = Original; Score experts: learned, (1 - G) x mixture + G x original (H=32, delta=0) = Score experts: learned weights per level (H=32)
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base. Median start: all quantiles start at the median (offsets 0).
Studentized full base: offsets start at the base; level RMS = the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

Single trackers
- Original
- Fixed MultiQT (eta=0.005*S0)
- Leaky MultiQT (eta=0.008*S0, leak 0.2)
- Delphi REF-op (sqrt, eta=0.03*Q90 of 20, slow 0.003, leak 0.1)
- Delphi flu prod (sqrt per 100k, eta=0.018)
- Count-slope rate (min m=0.1)
- Target oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=32)
- Rate grid: learned, no sharing, common weights (H=32)
- Rate grid: learned + sharing (H=32)
- Rate grid: learned + sharing, common weights (H=16)
- Rate grid: sharing + outer (H=32, outer m=0.001)
- Rate grid: sharing, common weights + outer (H=16, outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, H=32)
- Starts x rate grid: learned (H=32)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (H=32, outer m=0.001)
- Starts x rate grid + outer (H=32, outer m=0.001)

Score-based experts
- Decline MultiQT (eta=0.05*S0, declining context, delta=0)
- Score experts: equal weights
- Score experts: learned weights per level (H=32)
- Score experts: learned, (1 - G) x mixture + G x original (H=32, delta=0)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.03)
- Studentized median (RMS of 5, median start, eta=0.03)
- Studentized median (base 80% width, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, median start, eta=0.1)

Studentized full base
- Studentized full base (base 80% width, eta=0.1)
- Studentized full base (level RMS of 5, eta=0.03)

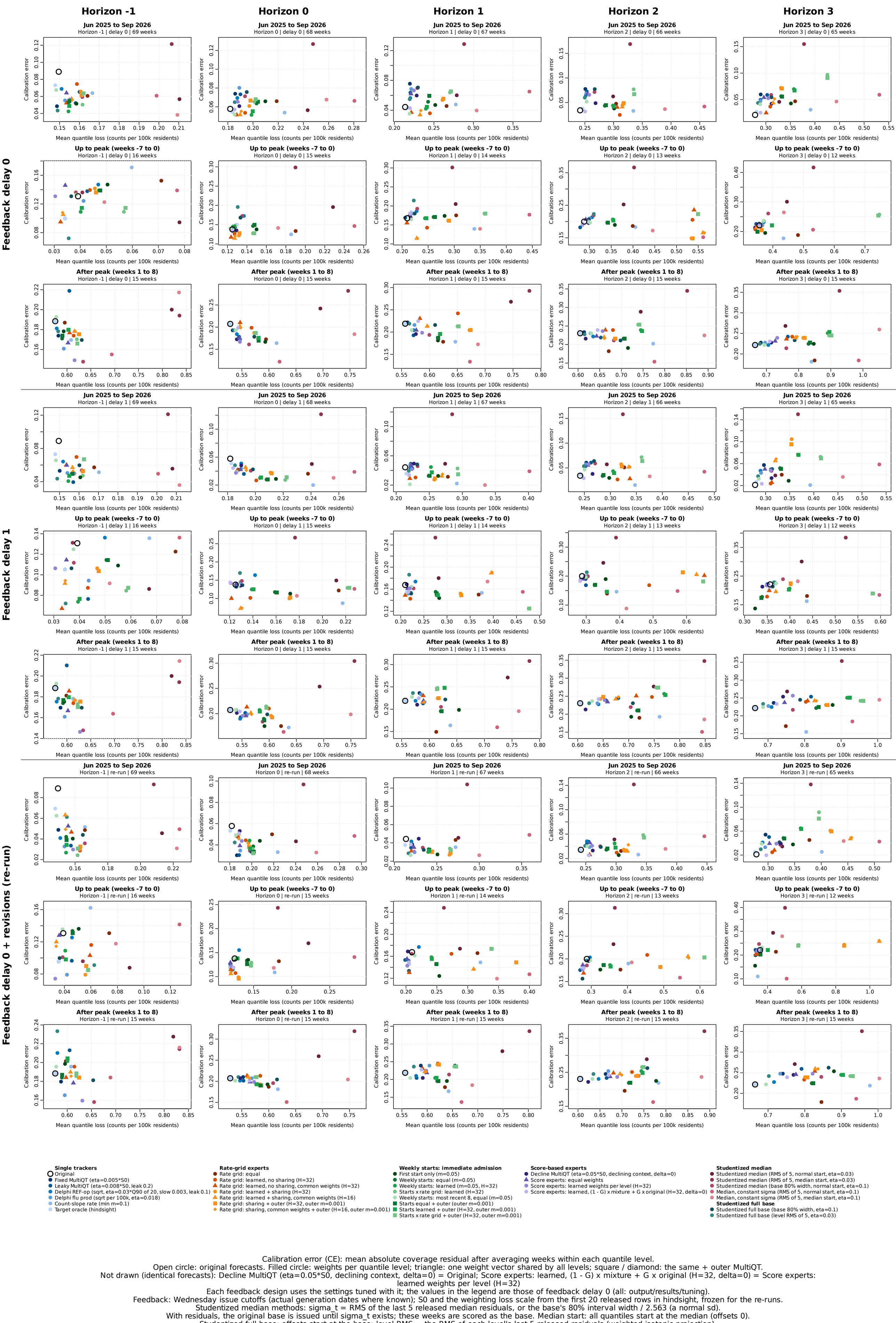
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Not drawn (identical forecasts): Decline MultiQT (eta=0.05*S0, declining context, delta=0) = Original; Score experts: learned, (1 - G) x mixture + G x original (H=32, delta=0) = Score experts: learned weights per level (H=32)
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base. Median start: all quantiles start at the median (offsets 0).
Studentized full base: offsets start at the base; level RMS = the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | New York (36) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

covid | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.



Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Not drawn (identical forecasts): Decline MultiQT (eta=0.05*S0, declining context, delta=0) = Original; Score experts: learned, (1 - G) x mixture + G x original (H=32, delta=0) = Score experts: learned weights per level (H=32)
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base. Median start: all quantiles start at the median (offsets 0).
Studentized full base: offsets start at the base; level RMS = the RMS of each level's last 5 released residuals (weighted isotonic projection).

flu | all 52 locations | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.
Small: single locations; large: means over locations; axes: 1st to 99th percentile of both.

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Not drawn (identical forecasts): Decline MultiQT (eta=0.05*S0, declining context, delta=0) = Original; Score experts: learned, (1 - G) x mixture + G x original (H=8, delta=0) = Score experts: learned weights per level (H=8)
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base. Median start: all quantiles start at the median (offsets 0).
Studentized full base: offsets start at the base; level RMS = the RMS of each level's last 5 released residuals (weighted isotonic projection).

flu | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Not drawn (identical forecasts): Decline MultiQT (eta=0.05*S0, declining context, delta=0) = Original; Score experts: learned, (1 - G) x mixture + G x original (H=8, delta=0) = Score experts: learned weights per level (H=8)
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base. Median start: all quantiles start at the median (offsets 0).
Studentized full base: offsets start at the base; level RMS = the RMS of each level's last 5 released residuals (weighted isotonic projection).

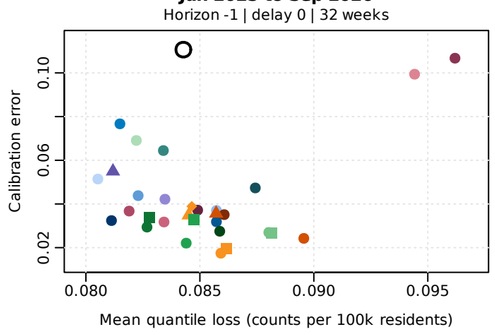
flu | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

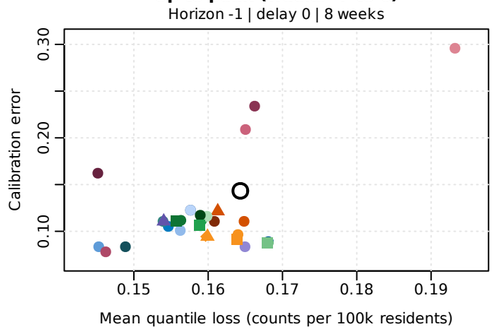
Horizon -1

Feedback delay 0

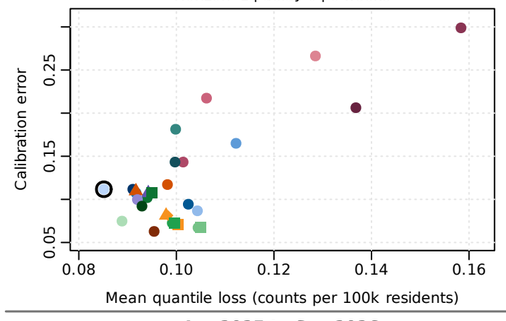
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 8 weeks

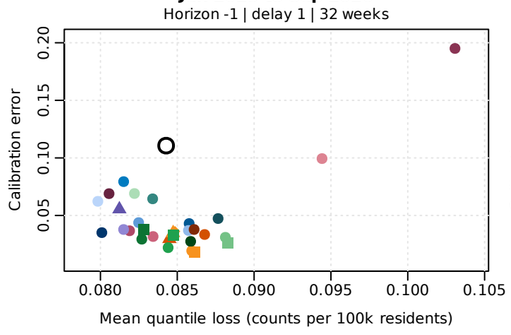


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 8 weeks

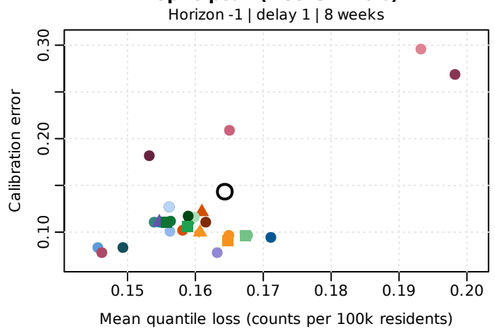


Feedback delay 1

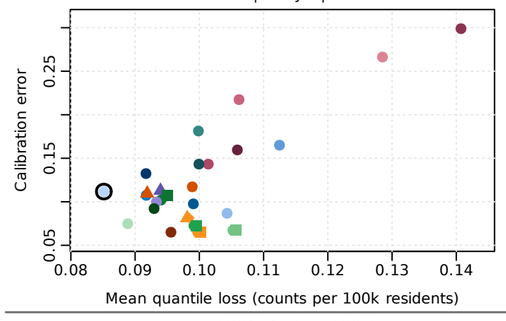
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 8 weeks

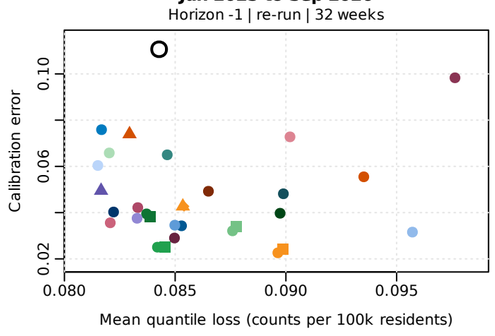


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 8 weeks

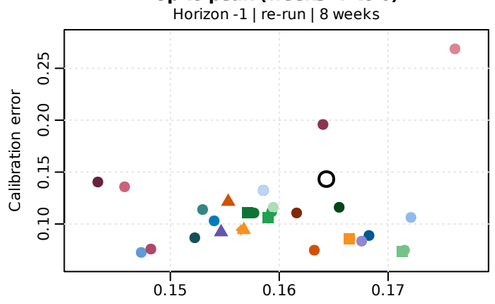


delay 0 + revisions (re-run)

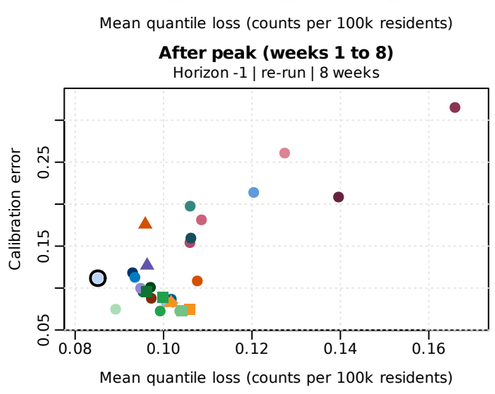
Jun 2025 to Sep 2026
Horizon -1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 8 weeks



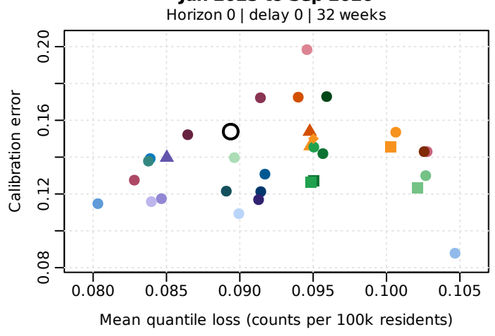
After peak (weeks 1 to 8)
Horizon -1 | re-run | 8 weeks



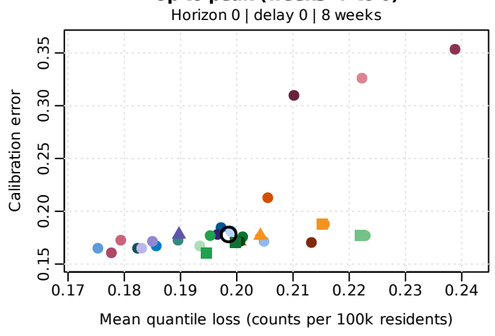
Horizon 0

Feedback delay 0

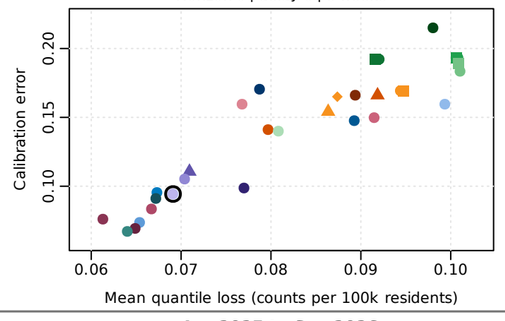
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 8 weeks

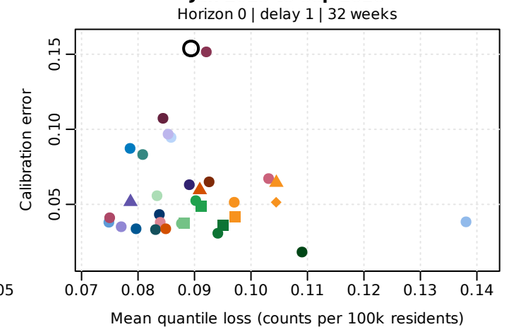


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 8 weeks

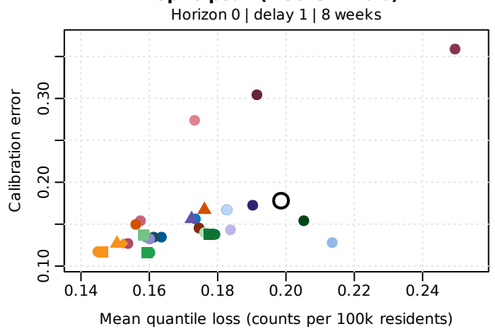


Feedback delay 1

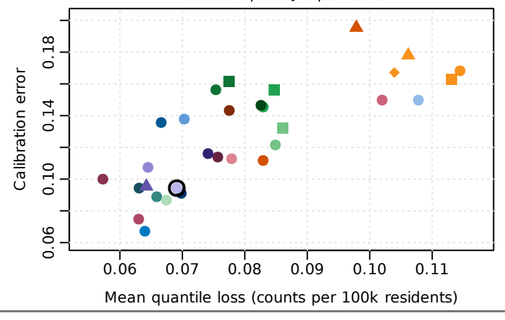
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 8 weeks

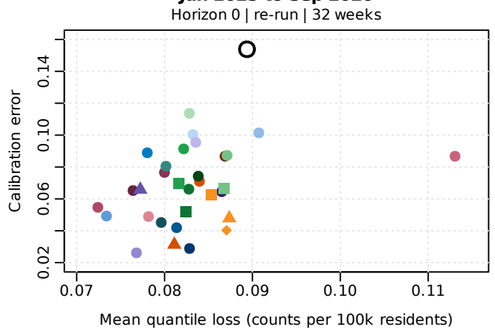


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 8 weeks

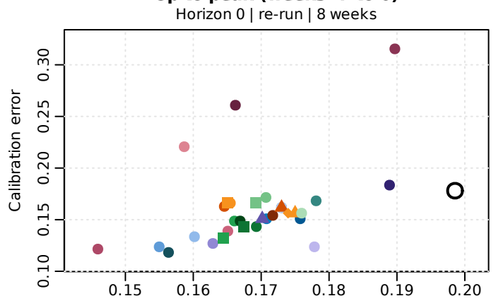


delay 0 + revisions (re-run)

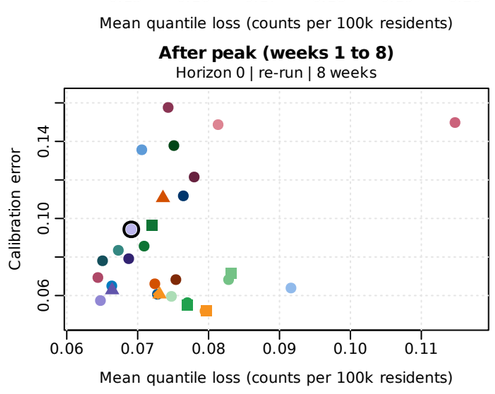
Jun 2025 to Sep 2026
Horizon 0 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 8 weeks



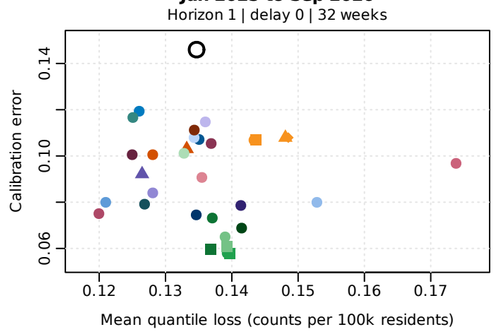
After peak (weeks 1 to 8)
Horizon 0 | re-run | 8 weeks



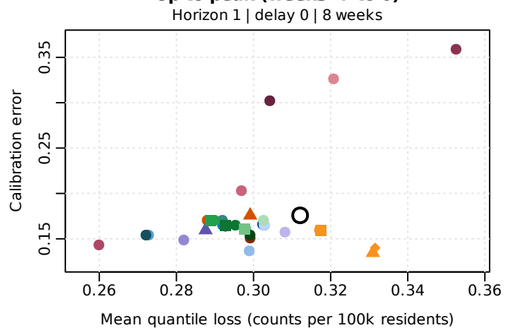
Horizon 1

Feedback delay 0

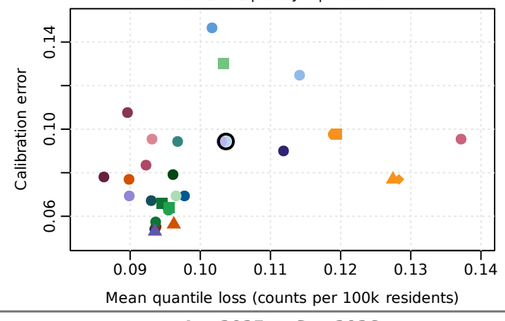
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 8 weeks

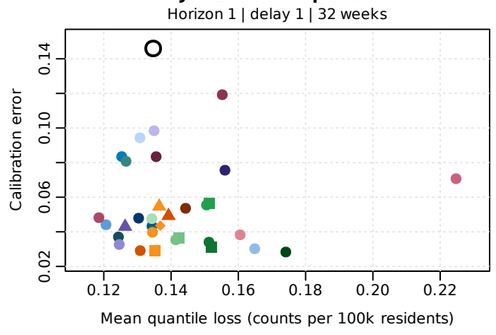


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 8 weeks

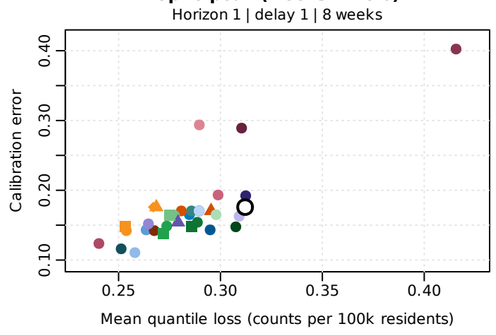


Feedback delay 1

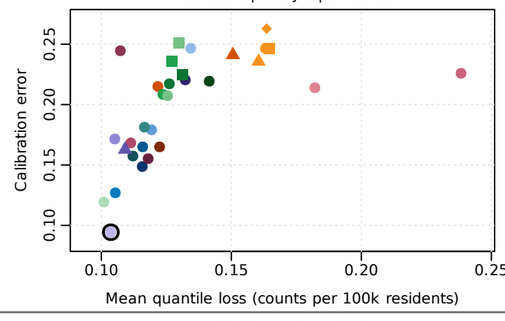
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 8 weeks

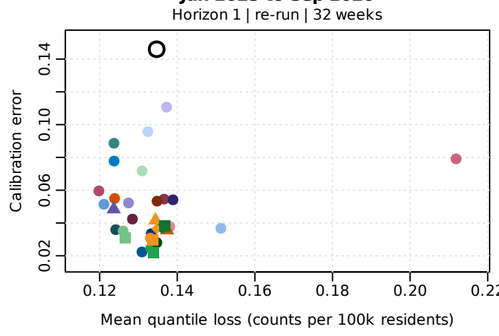


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 8 weeks

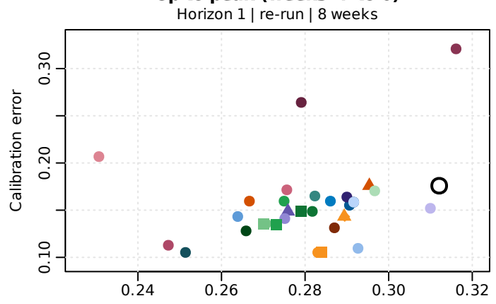


delay 0 + revisions (re-run)

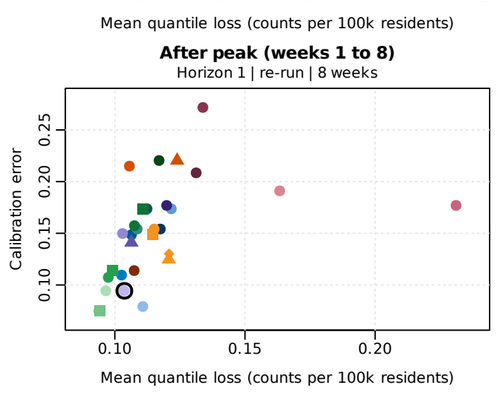
Jun 2025 to Sep 2026
Horizon 1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 8 weeks



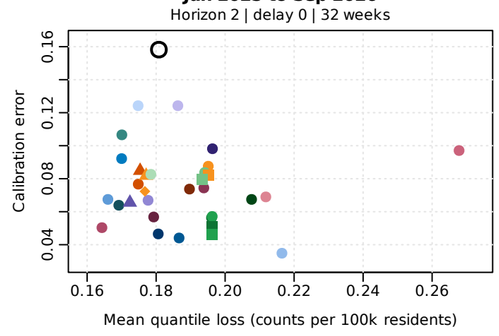
After peak (weeks 1 to 8)
Horizon 1 | re-run | 8 weeks



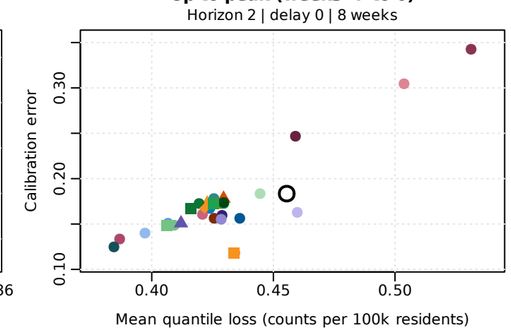
Horizon 2

Feedback delay 0

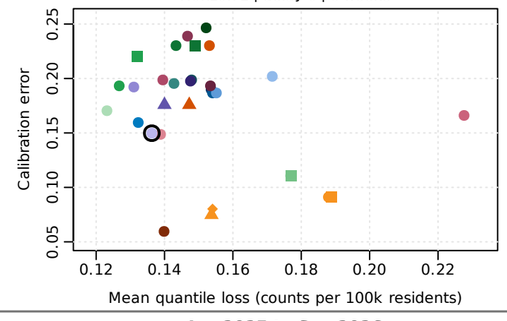
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 8 weeks

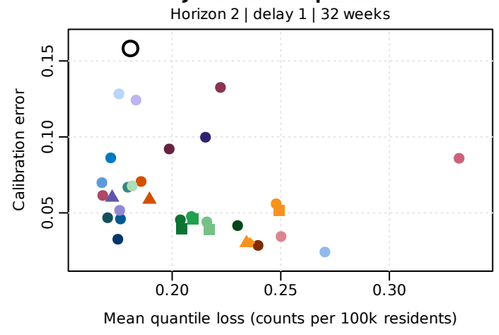


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 8 weeks

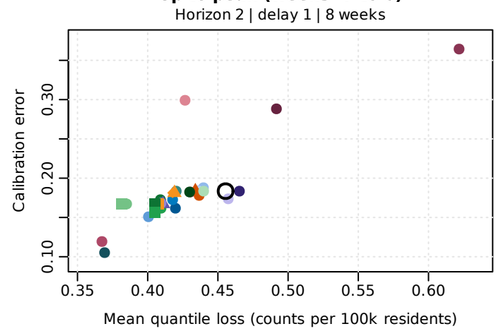


Feedback delay 1

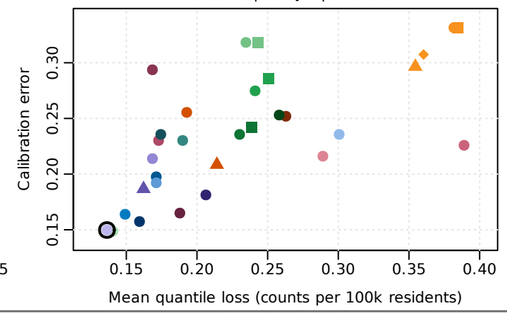
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 8 weeks

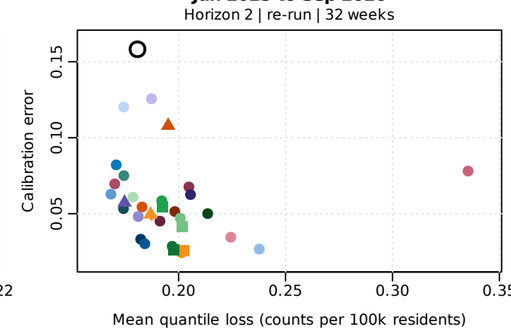


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 8 weeks

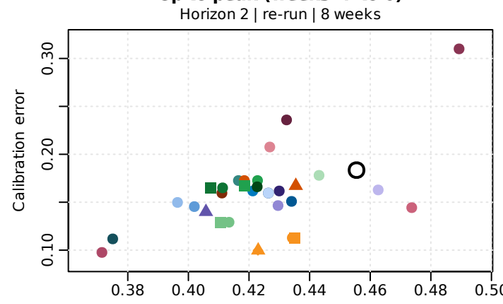


delay 0 + revisions (re-run)

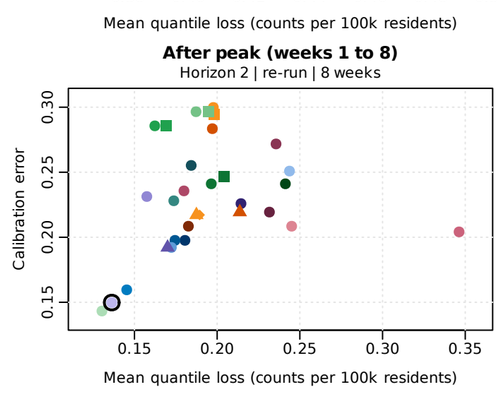
Jun 2025 to Sep 2026
Horizon 2 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 8 weeks



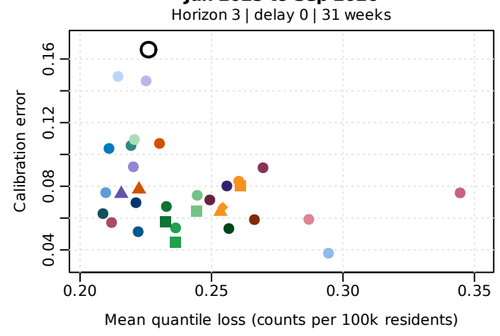
After peak (weeks 1 to 8)
Horizon 2 | re-run | 8 weeks



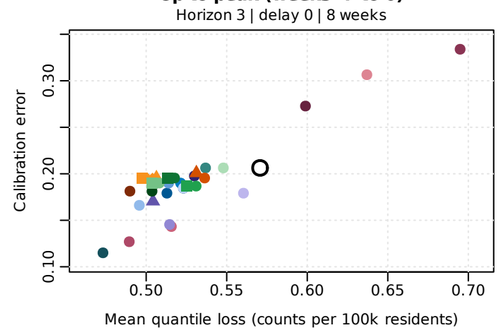
Horizon 3

Feedback delay 0

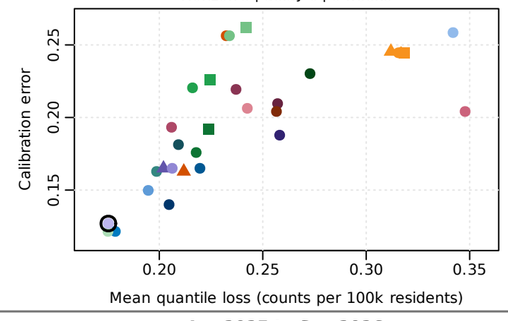
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 8 weeks

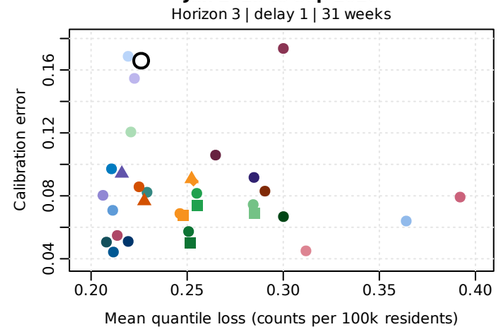


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 8 weeks

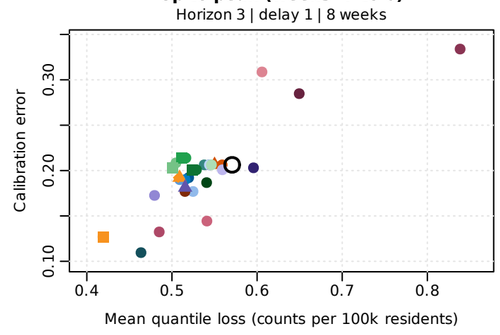


Feedback delay 1

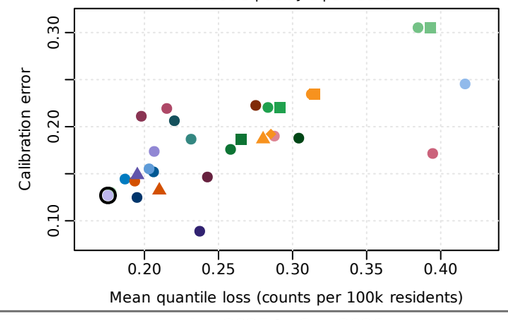
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 8 weeks

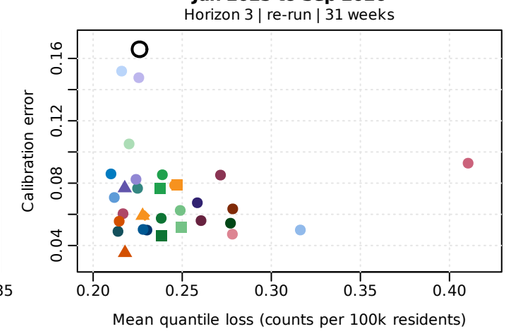


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 8 weeks

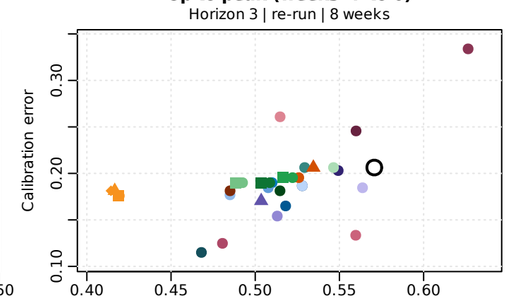


delay 0 + revisions (re-run)

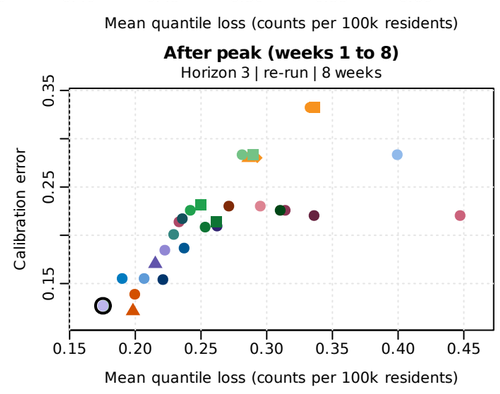
Jun 2025 to Sep 2026
Horizon 3 | re-run | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 8 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 8 weeks



Single trackers
- Original
- Fixed MultiQT (eta=0.02*S0)
- Leaky MultiQT (eta=0.02*S0, leak 0.2)
- Delphi REF-op (sqrt, eta=0.03*Q90 of 20, slow 0.003, leak 0.1)
- Delphi flu prod (sqrt per 100k, eta=0.018)
- Count-slope rate (min m=0.1)
- Target oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=2)
- Rate grid: learned, no sharing, common weights (H=16)
- Rate grid: learned + sharing (H=2)
- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (H=2, outer m=0.001)
- Rate grid: sharing, common weights + outer (H=2, outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, H=2)
- Starts x rate grid: learned (H=2)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (H=2, outer m=0.001)
- Starts x rate grid + outer (H=2, outer m=0.001)

Score-based experts
- Decline MultiQT (eta=0.05*S0, declining context, delta=0)
- Score experts: equal weights
- Score experts: learned weights per level (H=8)
- Score experts: learned, (1 - G) x mixture + G x original (H=8, delta=0)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.1)
- Studentized median (RMS of 5, median start, eta=0.1)
- Studentized median (base 80% width, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, median start, eta=0.1)

Studentized full base
- Studentized full base (base 80% width, eta=0.1)
- Studentized full base (level RMS of 5, eta=0.03)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Not drawn (identical forecasts): Decline MultiQT (eta=0.05*S0, declining context, delta=0) = Original; Score experts: learned, (1 - G) x mixture + G x original (H=8, delta=0) = Score experts: learned weights per level (H=8)
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base. Median start: all quantiles start at the median (offsets 0).
Studentized full base: offsets start at the base; level RMS = the RMS of each level's last 5 released residuals (weighted isotonic projection).

flu | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
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Studentized full base: offsets start at the base; level RMS = the RMS of each level's last 5 released residuals (weighted isotonic projection).

flu | New York (36) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

Single trackers
- Original
- Fixed MultiQT (eta=0.02*S0)
- Leaky MultiQT (eta=0.02*S0, leak 0.2)
- Delphi REF-op (sqrt, eta=0.03*Q90 of 20, slow 0.003, leak 0.1)
- Delphi flu prod (sqrt per 100k, eta=0.018)
- Count-slope rate (min m=0.1)
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- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (H=2, outer m=0.001)
- Rate grid: sharing, common weights + outer (H=2, outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, H=2)
- Starts x rate grid: learned (H=2)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (H=2, outer m=0.001)
- Starts x rate grid + outer (H=2, outer m=0.001)

Score-based experts
- Decline MultiQT (eta=0.05*S0, declining context, delta=0)
- Score experts: equal weights
- Score experts: learned weights per level (H=8)
- Score experts: learned, (1 - G) x mixture + G x original (H=8, delta=0)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.1)
- Studentized median (RMS of 5, median start, eta=0.1)
- Studentized median (base 80% width, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, median start, eta=0.1)

Studentized full base
- Studentized full base (base 80% width, eta=0.1)
- Studentized full base (level RMS of 5, eta=0.03)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
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Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base. Median start: all quantiles start at the median (offsets 0).
Studentized full base: offsets start at the base; level RMS = the RMS of each level's last 5 released residuals (weighted isotonic projection).

flu | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
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Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base. Median start: all quantiles start at the median (offsets 0).
Studentized full base: offsets start at the base; level RMS = the RMS of each level's last 5 released residuals (weighted isotonic projection).